Mod ① DE of 1st order & degree

Q2) $(y^2 e^{xy^2} + 4x^3)\cdot dx + (2xye^{xy^2} - 3y^2)\cdot dy = 0$

$u\cdot v = u\cdot v' + v\cdot u'$

$M = \underset{u}{y^2}\,\underset{v}{e^{xy^2}} + 4x^3$ $\quad N = \underline{2xy}_{u}\,\underline{e^{x^2y^2}}_{v} - 3y^2$

$$\boxed{\frac{\partial M}{\partial y} = y^2\cdot e^{xy^2}\cdot 2xy + 2y\cdot e^{xy^2}}$$

$$\frac{\partial N}{\partial x} = 2xy\cdot y^2 e^{xy^2}$$

↳ u did u·v rule wrong upon here so . . .

$$\boxed{\frac{\partial N}{\partial x} = 2xy\cdot e^{xy^2}\cdot y^2 + 2y\cdot e^{xy^2}}$$

as we can see $\boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}$ hence $\int M\cdot dx + \int N\cdot dy = 0$

↑ must be free from x

$$\int y^2 e^{xy^2} + 4x^3\cdot dx + \int -3y^2\cdot dy = 0$$

→ how to do this part?

$$\int y^2\cdot e^{xy^2} + \frac{\cancel{4}x^4}{\cancel{4}} + \frac{-\cancel{3}y^3}{\cancel{3}}$$

Q2) $\frac{y}{x^2}\cos\left(\frac{y}{x}\right)\cdot dx - \frac{1}{x}\cos\left(\frac{y}{x}\right)\cdot dy + 2x\cdot dx = 0$

$$\left(\frac{y}{x^2}\cdot\cos\left(\frac{y}{x}\right) + 2x\right)\cdot dx - \frac{1}{x}\cos\left(\frac{y}{x}\right)\cdot dy = 0.$$

$$M = \left(\underbrace{\frac{y}{x^2}}_{u}\cdot\underbrace{\cos\left(\frac{y}{x}\right)}_{v} + 2x\right)\cdot dx \qquad N = \left(\underbrace{-\frac{1}{x}}_{u}\underbrace{\cos\left(\frac{y}{x}\right)}_{v}\right)\cdot dy$$

$$\frac{\partial M}{\partial y} = \frac{y}{x^2}\left(-\sin\left(\frac{y}{x}\right)\cdot\frac{1}{x}\right) + \frac{1}{x^2}\cos\left(\frac{y}{x}\right)$$

$$\frac{\partial M}{\partial y} = -\frac{y}{x^3}\sin\left(\frac{y}{x}\right) \qquad \frac{\partial N}{\partial x} = -\frac{1}{x}\left(-\sin\left(\frac{y}{x}\right)\cdot\frac{y}{x^2}\right) - \frac{1}{x^2}\cos\left(\frac{y}{x}\right).$$

$$\frac{\partial N}{\partial x} = \frac{y}{x^3}\left(-\sin\left(\frac{y}{x}\right)\right) - \frac{1}{x^2}\cos\left(\frac{y}{x}\right).$$

Q2) Solve the following D.E

1. $(\underbrace{y^2e^{xy^2} + 4x^3}) \cdot dx + (2xy\, e^{xy^2} - 3y^2)\cdot dy = 0.$

this is of the format $Mdx + Ndy = 0$

$$M = y^2 e^{xy^2} + 4x^3 \qquad N = \underbrace{2xy}_{u}\,\underbrace{e^{xy^2}}_{v} - 3y^2$$

$$\frac{\partial M}{\partial y} = y^2\cdot 2xy\,e^{xy^2} + 2y\cdot e^{xy^2} \qquad \frac{\partial N}{\partial x} = 2xy\cdot y^2 e^{xy^2} + 2y\,e^{xy^2}$$

$$\frac{\partial M}{\partial y} = 2xy^3 e^{xy^2} + 2ye^{xy^2} \qquad \frac{\partial N}{\partial x} = 2xy^3 e^{xy^2} + 2ye^{xy^2}$$

$$\boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}$$

now the formula for GS $= \int M \cdot dx + \int N \cdot dy + c$.

$$\int y^2 e^{xy^2} + 4x^3 \cdot dx + \int -3y^2 \cdot dy + C$$

$$\int y^2 e^{xy^2} \Rightarrow \frac{1}{\cancel{y^2}} e^{xy^2} \cancel{y^2} - \frac{\cancel{3} y^3}{\cancel{3}}$$

$$\Rightarrow \underline{\underline{e^{xy^2} - y^3}}.$$

(2) $\frac{y}{x^2} \cos\left(\frac{y}{x}\right) \cdot dx - \frac{1}{x} \cos\left(\frac{y}{x}\right) \cdot dy + 2x \cdot dx = 0.$

$$\frac{y}{x^2} \cos\left(\frac{y}{x}\right) \cdot dx + 2x \cdot dx - \frac{1}{x} \cos\left(\frac{y}{x}\right) \cdot dy = 0$$

$$\left(\frac{y}{x^2} \cos\left(\frac{y}{x}\right) + 2x\right) \cdot dx - \frac{1}{x} \cos\left(\frac{y}{x}\right) \cdot dy = 0.$$

$$M\,dx + N\,dy = 0$$

$$M = \underbrace{\frac{y}{x^2}}_{u} \underbrace{\cos\left(\frac{y}{x}\right)}_{v} + 2x \qquad N = \underbrace{-\frac{1}{x}}_{u} \underbrace{\cos\left(\frac{y}{x}\right)}_{v}.$$

$$\frac{\partial M}{\partial y} = \underbrace{\frac{y}{x^2} \sin\left(\frac{y}{x}\right) \cdot \frac{1}{x}}_{} + \frac{1}{x^2} \cos\left(\frac{y}{x}\right)$$

$$\frac{\partial N}{\partial x} = +\frac{1}{x} \sin\left(\frac{y}{x}\right) \cdot \frac{y}{x^2} + \frac{1}{x^2} \cos\left(\frac{y}{x}\right).$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

free from x.

The General Solution $\int M \cdot dx + \int N \cdot dy = 0$

$$\int \left(\frac{y}{x^2}\cos\left(\frac{y}{x}\right) + 2x\right) \cdot dx + \int 0 + C.$$

$$\int \frac{\cancel{2}x^2}{\cancel{2}} + \underbrace{\frac{y}{x^2}}_{u} \underbrace{\cos\left(\frac{y}{x}\right)}_{v} \cdot dx.$$

let $u = \frac{y}{x}$ $\quad ux = y$ $\quad du = \frac{dy}{u}.$

$u\,dx = dy$

$$\int \cancel{u} \cos(u) \frac{dy}{\cancel{u}} \Rightarrow \sin u \Rightarrow \sin\left(\frac{y}{x}\right)$$

$$x^2 - \sin\left(\frac{y}{x}\right).$$

③ $(\underbrace{x^4e^x - 2mxy^2}_{M}) \cdot dx + (\underbrace{2mx^2y}_{N}) \cdot dy = 0$

$M = x^4e^x - 2mxy^2 \quad N = 2mx^2y$

$\frac{\partial M}{\partial y} = -4mxy \quad \frac{\partial N}{\partial x} = 4mxy$

$$\boxed{\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}}$$

using rule ③ $\dfrac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x)$ $\quad \therefore IF = e^{\int f(x) \cdot dx}.$

$\Rightarrow \dfrac{-\cancel{8}^{2}\cancel{mxy}}{\cancel{4}\cancel{mx}\cancel{xy}} = -2 = f(x) \quad IF = e^{-2x+C}$

multiply IF to the Quest.

$$\left((x^4 e^{x} - 2mxy^2)\cdot dx + (2mx^2 y)\cdot dy \right) e^{-2x}$$

$$x^4 e^{x}. e^{-2x} - 2mxy^2 . e^{-2x} + \left(2mx^2 y . e^{-2x}\right) \cdot dy$$

now the equation is Exact.

$$\Rightarrow GS = \int M\cdot dx + \cancel{\int N\cdot dy} \xrightarrow{} 0 + c.$$

let $2x = t$.

$2dx = dt$.

$$\int \underset{u}{x^4} \underset{v}{e^{-x}}.dx - \int 2mxy^2 . e^{-2x}. dx$$

$$v\cdot u' + u'' \int v + u''' \int v + u^{IV} \int v$$

$$\Rightarrow e^{-x} 4x^3 + 12x^2 e^{-x} + 24xe^{-x} + 24 e^{-x} + e^{x} x^4$$

+ by using integration by parts formula

$$\int dv = \int e^{-2x}.dx \Rightarrow v = -\frac{1}{2} e^{-2x} x$$

$$2my^2 \int (xe^{-2x}) \cdot dx = x\left(-\frac{1}{2} e^{-2x}\right) - \int -\frac{1}{2} e^{-2x}$$

$$2my^2 \left(x - \frac{1}{2} e^{-2x} - \frac{1}{4} e^{-2x} \right).$$

Q)3) i) $(x^2-1)\sin x \cdot \frac{dy}{dx} + [2x\sin x + (x^2-1)\cos x] y = (x^2-1)\cos x.$

this is of the type $\frac{dy}{dx} + Py = Qx.$

Dividing throuout $(x^2-1)\sin x$

$$\frac{dy}{dx} + \left[\frac{2x}{(x^2-1)} + \cot x\right] y = \cot x.$$

$$IF = e^{\int P dx}.$$

$$IF = e^{\int \left(\frac{2x}{x^2-1}\right) + \int (\cot x).dx}.$$

let $u = x^2 - 1$.

$du = 2x\,dx$.

$$= e^{\int \frac{du}{u} + \log \sin x}.$$

$$= e^{\log|u| + \log \sin x} = e^{\log|x^2-1| + \log \sin x}.$$

$$y.e^{\log|x^2-1| + \log \sin x} = \int e^{\log|x^2-1| + \log \sin x} \times$$

$$[2x \sin x + (x^2-1)\cos x].dx.$$

$$yIF = \int IF \cdot Q \cdot dx + c.$$

$$[x^2-1 + \sin x \cdot \times 2x \sin x + (x^2-1)\cos x].dx.$$

$$[2x^3 \sin x - 2x \sin x + 2x \sin^2 x + x^2 \cos x - \cos x] \cdot dx.$$

$$(-2x^3 \cos x - 6x^2 \sin x - 12x \cos x - 12 \sin x) - (2x(-\cos x) - 2(\sin x))$$

So on & so forth.

② LDE with constant coefficent of highest order

① $(D^2 - 2D - 4)y = 0$

$D^2 - 2D - 4 = 0$

$D = 1+\sqrt{5},\ 1-\sqrt{5}$

formula $y = C_1 e^{(1+\sqrt{5})x} + C_2 e^{(1-\sqrt{5})x}$

② $\frac{d^4y}{dx^4} + K^4 y = 0$

$D^4 y + K^4 y = 0$

$(D^4 + K^4) = 0$

$D^4 + K^4 = 0$

$(D = -K)$

③ $(D^2+1)^3 \underbrace{(D^2+D+1)^2}\, y = 0$

$D = (\pm i)^3 \qquad D = \frac{-1}{2} + \frac{\sqrt{3}i}{2},\ \frac{-1}{2} - \frac{\sqrt{3}i}{2}$

$= ((C_1 + C_2x + C_3x^2)\cos(x) + (C_4 + C_5x + C_6x^2)\sin x.$

Ⓘ $\left[(C_7 + C_8x)\cos\left(\frac{\sqrt{3}x}{2}\right) + (C_9 + C_{10}x)\sin\left(\frac{\sqrt{3}x}{2}\right)\right] \cdot e^{-\frac{1}{2}x}.$

$((c_1 + c_2x + c_3x^2)\cos(x) + (c_4 + c_5x + c_6x^2)\sin x +$

$\left[(c_7 + c_8x)\cos\left(\frac{\sqrt{3}x}{2}\right) + (c_9 + c_{10}x)\sin\left(\frac{\sqrt{3}x}{2}\right)\right] \cdot e^{-1/2x}$.

(4) $(D-1)^4(D^2+2D+2)^2 y = 0$

$\boxed{D=1}$ $D = -1+i, \; -1-i$

$\alpha \quad \beta$

$\Rightarrow$ for real & diffrent $y = c_1e^{m_1x} + c_2e^{m_2x} + \dots c_ne^{m_nx}$.

$y = c_1e^x + e^{-x}[c_2\cos x + c_3\sin(x)]$

(5) $(D^4 - 4D^3 + 8D^2 - 8D + 4)y = 0$.

$D = 1+i, 1-i, 1+i, 1-i \quad \Rightarrow (1+i) \quad (1-i)$

$\alpha = 1 \quad \beta = 1$

the formula for the imaginary & repeated.

$y = [(c_1x + c_2)\cos\beta x + ((c_3x + c_4)\cos\beta x]\, e^{\alpha x}$.

$y = [((c_1x + c_2)\cos x + (c_3x + c_4)\cos x]\, e^{x}$.

(6) $(D^2 - 7D - 6)y = (1+x^2)e^{2x}$.

$D^2 - 7D - 6 = 0 \quad D = 7.77, \; -0.77200$

now to 1st get the CF i.e Real & diffrent

$y = c_1e^{m_1x} + c_2e^{m_2x}. \Rightarrow c_1e^{7.77x} + c_2e^{(-0.772)x}$.

now the Type (I) States $\rightarrow e^{ax} \therefore PI = \frac{1}{f(D)}\cdot(1+x^2)e^{2x}$.

$$\frac{1}{(D^2-7D-6)}(1+x^2)e^{2x}.$$

$$\frac{1}{(D+2)^2-7(D+2)-6}(1+x^2)e^{2x}.$$

$$\frac{1}{D^2+4D+4-7D-14-6}(1+x^2)e^{2x} \Rightarrow \frac{1}{D^2-3D-16}(1+x^2)e^{2x}.$$

2 | 16
2 | 8
2 | 4
2
16

| 16

$$(D+2.77)(D-5.77).$$

$$\frac{1}{(D+2.77)(D-5.77)}(1+x^2)e^{2x}. \Rightarrow (1+x^2)e^{2x}.$$

fuck ts

Q1) Solve the following equations :-

① $(D^2 - 2D - 4)y = 0$

$D^2 - 2D - 4 = 0$

$x_1 = 3.23$; $x_2 = -1.236$

Both are real & diffrent.

$y_1 = c_1 e^{m_1 x} + c_2 e^{m_2 x}$

$$\boxed{y_1 = c_1 e^{3.23x} + c_2 e^{-1.236x}}$$

② $\frac{d^4y}{dx^4} + K^4 y = 0$

$D^4 y + K^4 y = 0$

$(D^4 + K^4)y = 0$

$D^4 = -K^4$

$D^4 = K^4 e^{i\pi + 2n\pi i}$ $\quad n = 0, 1, 2, 3$

$D = K e^{\frac{i\pi + 2n\pi i}{4}}$

if $n = 0$ $\quad K e^{\frac{i\pi + 2\pi i}{4}} \Rightarrow$

③ $\underbrace{(D^2+1)^3}\,\underbrace{(D^2+D+1)^2}\,y=0$

Solve these 2 seperately $\Rightarrow$ $D^2+1 = \pm i$

$D^2+D+1 = \underset{\alpha}{-\frac{1}{2}} \pm \underset{\beta}{\frac{\sqrt{3}i}{2}}$ $\Big\}$ each of them with multiplicity 3 & 2

In the case of Imaginary & same

$y = [(c_1 x + c_0)\cos\beta x + ((c_3 x + c_2)\sin\beta x] e^{\alpha x}$ → formula.

$y_1 = [((c_0 + c_1 x + c_2 x^2)\cos x + (c_3 + c_4 x + c_5 x^2)\sin x)]$

$y_2 = [((c_6 + c_7 x)\cos\frac{\sqrt{3}}{2}x + ((c_8 + c_9 x)\sin\frac{\sqrt{3}}{2}x] e^{-x/2}$

$$\boxed{\begin{aligned} y_1 \times y_2 = {} & [((c_0 + c_1 x + c_2 x^2)\cos x + (c_3 + c_4 x + c_5 x^2)\sin x)] \times \\ & [((c_6 + c_7 x)\cos\tfrac{\sqrt{3}}{2}x + ((c_8 + c_9 x)\sin\tfrac{\sqrt{3}}{2}x] e^{-x/2} \end{aligned}}$$

④ $\underbrace{(D-1)^4}_{y_1}\,\underbrace{(D^2+2D+2)^2}_{y_2}\,y=0$

$D = 1$ multiplicity with 4

$D^2+2D+2 = \underset{\alpha}{-1} \pm \underset{\beta}{i}$ multiplicity with 2

for real & repeated the formula $\Rightarrow$ $(c_1 x + c_2)e^{\alpha x}$

$(c_1 x^3 + c_2 x^2 + c_3 x + c_4)e^{x} = y_1$

$y_2 = [(a_1 + a_2 x)\cos\beta x + (a_3 + a_4 x)\sin\beta x]e^{\alpha x}$

$y_2 = ((a_1 + a_2 x)\cos x + (a_3 + a_4 x)\sin x]e^{-x}$

$y_1 + y_2 = (c_1 x^3 + c_2 x^2 + c_3 x + c_4)e^{x} +$

$((a_1 + a_2 x)\cos x + (a_3 + a_4 x)\sin x]e^{-x}$

⑤ $(D^4 - 4D^3 + 8D^2 - 8D + 4)y = 0$

$D = 1 \pm i$ Each with multiplicity 2

$\therefore y(x) = [(c_1 + c_2 x)\cos\beta x + (c_3 + c_4 x)\sin\beta x]e^{\alpha x}$

$[(c_1 + c_2 x)\cos x + (c_3 + c_4 x)\sin x]e^{x}$

Variation of parameters :-

① $\frac{d^2y}{dx^2} - y = \frac{2}{1+e^x}$

$D^2y - y = \frac{2}{1+e^x}$

$y(D^2 - 1) = \frac{2}{1+e^x}$

$\boxed{D = \pm 1}$

$CF \Rightarrow c_1 e^x + c_2 e^{-x}$

$y_1 = e^x \quad y_2 = e^{-x}$

$w = \begin{vmatrix} e^x & e^{-x} \\ e^x & -e^{-x} \end{vmatrix} \Rightarrow -e^x e^{-x} - e^{-x} \cdot e^x = -2$

$u = -\int \frac{y_2 X}{w} dx \qquad v = \int \frac{y_1 X}{w} dx.$

$X = \frac{2}{1+e^x}$

$$u = +\int \frac{e^{-x}}{2}\left(\frac{2}{1+e^x}\right) \qquad v = \int \frac{e^x}{-2}\left(\frac{2}{1+e^x}\right)$$

$$u = \int \frac{e^{-x}}{1+e^x} \qquad v = \int \frac{-e^x}{1+e^x}$$

let $t = e^x$

$dt = e^x dx \qquad \frac{dt}{t} = dx \quad ; \quad e^{-x} = \frac{1}{t}.$

$$u = \int \frac{1/t}{1+t} \cdot \frac{dt}{t} = \int \frac{dt}{t^2(1+t)}$$

$$\frac{1}{t^2(1+t)} = \frac{-1}{t} + \frac{1}{t^2} + \frac{1}{1+t}$$

$$u = -\ln|t| - \frac{1}{t} + \ln|1+t| = -x - e^{-x} + \ln(1+e^x)$$

$$v = -\int \frac{e^x}{1+e^x} dx = -\ln(1+e^x) + \text{const.}$$

$PI = uy_1 + vy_2$

$y = CF + PI$

$y = c_1 e^x + c_2 e^{-x} + uy_1 + vy_2$

(2) $\frac{d^2y}{dx^2} + y = \sec x \cdot \tan x$

$D^2y + y = \sec x \cdot \tan x \rightarrow X$

$(D^2+1) = X \qquad D = \pm i$

$\alpha \quad \beta$

$$CF = [c_1 \cos x + c_2 \sin x] e^{0x}$$

$$CF = c_1 \underbrace{\cos x}_{y_1} + c_2 \underbrace{\sin x}_{y_2}$$

$$y_1 = \cos x \quad ; \quad y_2 = \sin x$$

$$w = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

$$w = \cos^2 x + \sin^2 x = 1$$

$$w = 1$$

$$u = -\int \frac{y_2 \times}{w} dx. \qquad v = \int \frac{y_1 \times}{w} dx$$

$$u = -\int \frac{\sin x}{1} \sec x \cdot \tan x \qquad v = \int \frac{\cos x}{1} \sec x \cdot \tan x$$

$$u = \int \sin x \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} \qquad v = \int \cancel{\cos x} \cdot \frac{1}{\cancel{\cos x}} \tan x$$

$$u = \int \tan^2 x \qquad v = \int \tan x$$

$$u = \int (1 - \sec^2 x) dx \qquad v = -\log \cos x$$

$$u = x - \tan x$$

$$PI = u y_1 + v y_2$$

$$= (x - \tan x) \cos x + (-\log \cos x) \sin x$$

$$\boxed{y = x \cos x - \sin x - (\log \cos x) \sin x + c_1 \cos x + c_2 \sin x.}$$

(3) $\frac{d^2y}{dx^2} + y = \frac{1}{1+\sin x}$

$D^2y + y = \frac{1}{1+\sin x} \rightarrow X.$

$D = \pm i \Rightarrow \underbrace{c_1\cos x}_{y_1} + \underbrace{c_2\sin x}_{y_2} = CF.$

$y_1 = \cos x \qquad y_2 = \sin x$

$w = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix} = 1$

$u = -\int \sin x \frac{1}{1+\sin x}\,dx \qquad v = \int \frac{\cos x}{1+\sin x}\,dx$

$u = -\int \frac{\sin x}{1+\sin x}\,dx. \qquad$ Let $1+\sin x = t.$

$\cos x \cdot dx = dt.$

$u = \int \left(-1 + \frac{1}{1+\sin x}\right) \qquad v = \int \frac{dt}{t}$

$u = -x + \int \frac{dx}{1+\sin x} \qquad v = \int \frac{1}{t} = \ln(t)$

$\ln(1+\sin x)$

$\boxed{\frac{1}{1+\sin x} = \tan x - \sec x}$

$\ln(1+\sin x)$

$u = -x + \tan x - \sec x$

$y_1u + y_2v \Rightarrow \cos x(-x + \tan x - \sec x) + \sin x(\ln(1+\sin x))$

$y = CF + PI = c_1\cos x + c_2\sin x + \cos x(-x + \tan x - \sec x) + s\,(\ln(1+\sin x))$

(4) $(D^2 - 6D + 9)y = \dfrac{e^{3x}}{x^2}$

$$X = \frac{e^{3x}}{x^2}$$

$D^2 - 6D + 9$

$D = 3, 3$ $\quad CF = (C_1 + C_2 x)e^{3x}$.

$$CF = \underbrace{C_1 e^{3x}}_{y_1} + \underbrace{C_2 x e^{3x}}_{y_2}$$

$y_1 = e^{3x} \qquad y_2 = xe^{3x}$

$$w = \begin{vmatrix} e^{3x} & x \cdot e^{3x} \\ 3e^{3x} & x \cdot 3e^{3x} + e^{3x} \end{vmatrix}$$

$\underset{u}{x} \cdot \underset{v}{e^{3x}} = v \cdot u' + u v'$

$x \cdot 3e^{3x} - e^{3x}$

$w = e^{3x}(+x \cdot e^{3x} + e^{3x}) - x \cdot e^{3x} e^{3x}$

$w = x\,\cancel{e^{6x}} - e^{6x} - \cancel{x \cdot e^{6x}}$

$w = +e^{6x}$

$u = -\displaystyle\int \frac{y_2 X}{w}\,dx \qquad v = \int \frac{y_1}{w}\,dx.$

$u = -\displaystyle\int \frac{\cancel{x e^{3x}}}{\cancel{e^{6x}}} \frac{e^{3x}}{x^2} = \int -\frac{1}{x} \qquad v = \int \frac{e^{3x}}{e^{6x}} \frac{e^{3x}}{x^2}$

$u = -\displaystyle\int (x)^{-1} \qquad v = \int \frac{1}{x^2}$

$u = -\ln|x| \qquad v = -1/x.$

Pg ③ Solve the following D.E

① $(y^2 e^{xy^2} + 4x^3)\,dx + (2xy e^{xy^2} - 3y^2)\,dy = 0$.

$(y^2 e^{xy^2} + 4x^3)\,dx = -(2xy e^{xy^2} - 3y^2).dy$

$\underbrace{(y^2 e^{xy^2} + 4x^3)}\,dx + \underbrace{(2xy e^{xy^2}} - 3y^2)\,dy = 0$

$M(x,y)\,dx \quad + \quad N(x,y)\,dy = 0$

$\frac{\partial M}{\partial y} = \underset{u}{y^2}\,\underset{v}{e^{xy^2}} + 4x^3 \qquad \frac{\partial N}{\partial x} = (\underset{u}{2xy}\,\underset{v}{e^{xy^2}} - 3y^2)$

$\frac{\partial M}{\partial y} = y^2 2xy e^{xy^2} + 2y e^{xy^2} \qquad \frac{\partial N}{\partial x} = 2xy\,(y^2)e^{xy^2} + 2y e^{xy^2}$

Since in this case $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

∴ the Solution'll be

$$\int M.dx + \int \left[N - \frac{\partial}{\partial x}\left(\int M\,dx\right)\right].dy = c$$

$$\int (y^2 e^{xy^2} + 4x^3).dx + \int \left\{(2xy e^{xy^2} - 3y^2) - \frac{\partial}{\partial x}\int (y^2 e^{xy^2} + 4x^3)\,dx\right\}dy$$

let $u = xy^2$

$\frac{du}{dx} = y^2 \qquad \int e^u.du + \int u x^3\,dx$.

$y^2 dx = du$

$e^u du + \frac{\cancel{4}x^4}{\cancel{4}}$

$= \boxed{e^{xy^2} + x^4}$

$$\int \left[\left(2xye^{xy^2} - 3y^2\right) - \frac{\partial}{\partial x}\int \left(y^2 e^{xy^2} + 4x^3\right) dx \right] dy$$

let $xy^2 = u$

$y^2 \cdot dx = du$

$$\left[\frac{\partial}{\partial x} \int du \cdot e^u + \int 4x^3 \cdot dx \right]$$

$$\frac{\partial}{\partial x}\left[e^u + x^4 \right]$$

$\frac{\partial}{\partial x} \rightarrow$ treat y as constant.

$$\frac{\partial}{\partial x}\left[e^{xy^2} + x^4 \right] \Rightarrow y^2 \cdot e^{xy^2} + 4x^3$$

$$\int \left[\left(2xye^{xy^2} - 3y^2\right) - y^2 e^{xy^2} - 4x^3 \right] . dy$$

That was an wayy more chaotic approach rather were gonna use

$\int M \cdot dx + \int N \cdot dy + C$ → This should be free from x.

Since in this case it isn't the answer'll be

$$\boxed{e^{xy^2} + x^4}$$

② $\frac{y}{x^2} \cos\left(\frac{y}{x}\right) dx - \frac{1}{x} \cos\left(\frac{y}{x}\right) dy + 2x\, dx = 0$

$$\Rightarrow \underbrace{\left(\frac{y}{x^2} \cos\left(\frac{y}{x}\right) + 2x \right)}_{M} dx - \underbrace{\frac{1}{x} \cos\left(\frac{y}{x}\right)}_{N} . dy = 0$$

$$\Rightarrow \quad \frac{\partial M}{\partial y} = \left(\underbrace{\frac{y}{x^2}}_{u} \underbrace{\cos\left(\frac{y}{x}\right)}_{v} + 2x \right) \qquad \frac{\partial N}{\partial x} = \left(\underbrace{\frac{1}{x}}_{u} \underbrace{\cos\left(\frac{y}{x}\right)}_{v^2} \right)$$

$$\frac{\partial M}{\partial y} = \frac{y}{x^2} \cdot \sin\left(\frac{y}{x}\right)\left(\frac{1}{x}\right) + \frac{1}{x^2}\sin\left(\frac{y}{x}\right) \qquad \frac{\partial N}{\partial x} : \frac{1}{x}\sin\left(\frac{y}{x}\right)\frac{y}{x^2} + \frac{1}{x^2}\sin\left(\frac{y}{x}\right)$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

the general soln $\int M \cdot dx + \underbrace{\int N \cdot dy}$ This is free from x.

$$\int\left(\frac{y}{x^2}\cos\left(\frac{y}{x}\right) + 2x\right)dx$$

$\frac{y}{x} = u$

$$-\frac{1}{2}\int \cos(u) \cdot du + \int 2x \cdot dx$$

$-2yx^{-2}dx = du$

$$-\frac{1}{2}\sin(u) \cdot \quad + \frac{\cancel{2}x^2}{\cancel{2}}$$

$-\frac{2y}{x^2}dx = dy$

$\frac{y}{x^2}dx = -\frac{1}{2}dy$

$$= \boxed{-\frac{1}{2}\sin\left(\frac{y}{x}\right) + x^2}$$

③

Mod ③ Γ & B f(n) ; nx^{n-1} ↓ ∫

Duis

This should be of the form $\boxed{\int_0^\infty e^{-x} x^{n-1} \cdot dx = \Gamma n}$

$$\int_0^\infty e^{-x} x^n \cdot dx = \Gamma n+1$$

(a) $\int_0^\infty (x^2+4)\, e^{-2x^2} \cdot dx$. Let $2x^2 = t$.

$x = \sqrt{\frac{t}{2}}$.

$$\int_0^\infty x^2 e^{-2x^2} dx + \int_0^\infty e^{-2x^2} \cdot 4\, dx$$

$dx = \frac{1}{\sqrt{2}} \frac{1}{2\sqrt{t}}$

$(t)^{1/2 - 1}$

$-\frac{1}{2} \frac{1}{\sqrt{t}}$.

$\boxed{dx = \frac{-1}{2\sqrt{2t}}}$

$$\int_0^\infty \frac{t}{2} e^{-t} \frac{-1}{2\sqrt{2t}} + 4\int_0^\infty e^{-2x^2} \frac{1}{2\sqrt{2t}}.$$

$$-\frac{1}{2\sqrt{2}} \int_0^\infty t^{1/2} e^{-t} + \frac{\cancel{2} \times \sqrt{2}}{\sqrt{2} \times \sqrt{2}} \int_0^\infty e^{-2x^2} t^{-1/2} \cdot dt.$$

$$-\frac{1}{2\sqrt{2}} \int_0^\infty e^{-t} t^{3/2 - 1} \cdot dt + \sqrt{2} \int_0^\infty e^{-t} t^{1/2 - 1}$$

$\frac{1}{2} = n - 1$

$$-\frac{1}{2\sqrt{2}}\Gamma\left(\frac{3}{2}\right) + \sqrt{2}\,\Gamma\left(\frac{1}{2}\right)$$

$$-\frac{1}{2\sqrt{2}}\,\frac{1}{2}\Gamma\left(\frac{1}{2}\right) + \sqrt{2}\sqrt{\pi}$$

$$-\frac{1}{2\sqrt{2}}\,\frac{1}{2}\sqrt{\pi} + \sqrt{2}\sqrt{\pi}$$

$$-\frac{1}{4\sqrt{2}}\sqrt{\pi} + \frac{\sqrt{2}\cdot 4\sqrt{2}}{4\sqrt{2}}\sqrt{\pi} = \frac{1}{4\sqrt{2}}\sqrt{\pi}\left[-1+8\right]$$

$$= \boxed{\frac{7\sqrt{\pi}}{4\sqrt{2}}}$$

(b) $\displaystyle\int_0^1 \frac{dx}{\sqrt{-\log x}}$

formula for β $f(x) = \displaystyle\int_0^1 x^{m-1}(1-x)^{n-1}\,dx = \beta(m,n)$

(c) $\int_0^\infty x^2\, 7^{-4x^2}\, dx$.

(1) The trouble maker $7^{-4x^2} = e^{-4x^2 \ln 7}$.

let $a = +4\ln 7$.

$$I = \int_0^\infty x^2 e^{-ax^2} \cdot dx.$$

let $t = ax^2$ $\quad x = \sqrt{\frac{t}{a}}$ $\quad \therefore dx = \frac{1}{2\sqrt{at}} \cdot dt.$

$x^2 = t/a$. $\quad I = \int_0^\infty \frac{t}{a} e^{-t} \cdot \frac{1}{2\sqrt{at}} \cdot dt$

$$I = \frac{1}{2a\sqrt{a}} \int_0^\infty t \cdot (t)^{-1/2} e^{-t} \cdot dt.$$

$$I = \frac{1}{2a\sqrt{a}} \int_0^\infty (t)^{1/2} e^{-t} \cdot dt.$$

the format should be $I = \int_0^\infty e^{-t} \cdot t^{n-1} \cdot dt$ $\longrightarrow$ for this $\underline{\underline{n = 3/2}}$.

Alternatively this formula is also valid

$$? = \int_0^\infty e^{-t}\, t^n = \Gamma(n+1)$$

$$\frac{1}{2a\sqrt{a}} \Gamma\frac{3}{2} = \boxed{\frac{\sqrt{\pi}}{4a\sqrt{a}}}$$ ← u can solve further if you want.

ⓑ $\int_0^1 \frac{dx}{\sqrt{-\log x}}$

let $t = -\log x$ $\therefore x = e^{-t}$.

$$\boxed{dx = -e^{-t}.dt.}$$

when $x = 0 : t = \infty$

$x = 1$ $t = 0$

So the new range $= (\infty, 0)$

$$\therefore \int_{t=\infty}^{0} \frac{-e^{-t}.dt}{\sqrt{t}}.$$

of the form

$$= \int_0^\infty (t)^{-1/2} e^{-t} \cdot dt. \Rightarrow \int_0^\infty e^{-t} t^{n-1} dt = \Gamma n$$

$$\Gamma\frac{1}{2} = \sqrt{\pi}.$$

ⓓ $\int_0^\infty 3^{-4x^2} dx.$

let $t = e^{-4x^2 \ln 3}$

let $a = +4\ln 3$

now e^{-ax^2}

$$\int_0^\infty e^{-ax^2} dx$$

$$\int_0^\infty x^0 e^{-ax^2}\,dx.$$

Let $t = ax^2$ $\qquad \sqrt{\frac{t}{a}} = x$

$$dx = \frac{1}{2\sqrt{t\cdot a}}\cdot dt.$$

$$\int_0^\infty \left(\sqrt{\frac{t}{a}}\right)^0 \cdot e^{-t}\,\frac{1}{2\sqrt{t\cdot a}}\cdot dt.$$

Formula $\int_0^\infty x^n e^{-x}\,dx = \Gamma{n+1}$

$$\frac{1}{2\sqrt{a}}\int_0^\infty e^{-t}\,(t)^{-1/2}\cdot dt$$

$$\frac{1}{2\sqrt{a}}\,\Gamma{-\frac{1}{2}+1} = \frac{1}{2\sqrt{a}}\,\Gamma{\frac{1}{2}} = \frac{1\times\sqrt{\pi}}{2\sqrt{a}}.$$

$$\frac{\sqrt{\pi}}{2\times\sqrt{4\ln 3}} \qquad \boxed{\frac{\sqrt{\pi}}{4\sqrt{\ln 3}}}$$

(e) $\int_0^\infty \frac{x^2}{(1+x^6)^{5/2}}\cdot dx.$

Step (1) Proper Substitution $\quad x^6 = (x^2)^3$

Standard trick $\quad t = x^3 \,;\; dt = 3x^2\,dx \quad$ i.e $\quad x^2\,dx = dt/3$

$$\int_0^\infty (1+t^2)^{5/2}\cdot\frac{dt}{3}.$$

Let $t = \tan\theta$. So the limit changes to $\int_0^{\pi/2}$

$$\int_0^{\infty} \sec^5\theta \cdot \frac{dt}{3} \Rightarrow \int_0^{\pi/2} \frac{\sec^2\theta \cdot d\theta}{\sec^5\theta} \Rightarrow \int_0^{\pi/2} \cos^3\theta \cdot d\theta$$

$$\int_0^{\pi/2} \cos^3\theta \cdot d\theta$$

$$\cos^3\theta = \cos\theta(1 - \sin^2\theta)$$

$$\boxed{\int_0^{\pi/2} \cos^n\theta \, d\theta = \frac{\sqrt{\pi}}{2} \frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2}+1\right)}}$$

$$\int_0^{\pi/2} \cos^3\theta \cdot d\theta : \frac{\sqrt{\pi}}{2} \frac{\Gamma\left(\frac{3+1}{2}\right)}{\Gamma\left(\frac{3}{2}+1\right)}$$

$$= \frac{\sqrt{\pi}}{2} \frac{\Gamma(2)}{\Gamma\left(\frac{5}{2}\right)} = \frac{\sqrt{\pi}}{2} \frac{1}{\frac{3}{2} \cdot \frac{1}{2}\Gamma\frac{1}{2}}$$

$$= \frac{\cancel{\sqrt{\pi}}}{2} \frac{1}{\frac{3}{4}\cancel{\sqrt{\pi}}} = \frac{2}{3}$$

$$I = \frac{1}{3} \times \frac{2}{3} = \boxed{\frac{2}{9}}$$

(f) $$\int_0^{\infty} \frac{x^3(x^3-1)}{(1+x^3)^5} x^2 \cdot dx$$

Let $x^3 = t$ $\qquad$ $x^3 = t.$ ✓ feels right.

$x = t^{1/3}$ $\qquad$ $3x^2 dx = dt.$

$dx = \frac{1}{3} t^{-2/3} . dt.$ $\qquad$ $x^2 dx = \frac{dt}{3}$

$$\int_0^\infty \frac{t(t-1)}{(1+t^3)^5}\,\frac{dt}{3}$$

let $u = \frac{1}{1+t}$. so $t = \frac{1-u}{u}$, $dt = \frac{-du}{u^2}$

$t=0$: $u=1$ when $t=\infty$ $u \to 0^+$

$1+t = \frac{1}{u}$ so $(1+t^5) = u^{-5}$

& $t-1 = \frac{1-u}{u} - 1 = \frac{1-u-u}{u} = \frac{1-2u}{u}$

$t = \frac{1-u}{u}$

$t(t-1) = \frac{1-u}{u} \cdot \frac{1-2u}{u} = \frac{(1-u)(1-2u)}{u^2}$

now $\frac{t(t-1)}{(1+t)^5} = \frac{(1-u)(1-2u)}{u^2} \cdot u^5 = (1-u)(1-2u)u^3$

also $dt = \frac{-du}{u^2}$

$$I = \frac{1}{3}\int_{u=1}^{0} (1-u)(1-2u)u^3 \cdot \left(\frac{-du}{u^2}\right)$$

$$I = \frac{1}{3}\int_0^1 (1-u)(1-2u)u^{3-2} \cdot du$$

$$I = \frac{1}{3}\int_0^1 (1-u)(1-2u)u \cdot du$$

Integrate & substitute

$$I = \frac{1}{3}\cdot 0 = 0 \qquad \boxed{0}$$

(a) $$\int_3^7 \sqrt[4]{(x-3)(7-x)}\, dx.$$

Step ① firstly we need to shift the symmetry

let $t = x-5 \quad (3 \xrightarrow{2} 5 \xrightarrow{2} 7)$

$x = t+5 \; ; \; dx = dt.$

x	t
3	-2
7	2

also $(x-3) = t+2$

$(7-x) = 2-t$

$\therefore \; (x-3)(7-x) = (t+2)(2-t) = 4-t^2$

$\therefore \; I = \int_{-2}^{2} (4-t^2)^{1/4}\, dt.$ ← This is an even function.

so if there's an even function which is $f(-t) = f(t)$ then when u integrate that fn over an symmetric interval $[-a, a]$ the area from $[-a, 0]$ to $[0, a]$ is exactly the same.

$$\therefore \; I = 2\int_0^2 (4-t^2)^{1/4}\, dt.$$

let $t = 2\sin\theta$

$dt = 2\cos\theta \cdot d\theta.$

t	θ
0	0
2	$\pi/2$

$4-t^2 = 4 - 4\sin^2\theta = 4\cos^2\theta$

$(4-t^2)^{1/4} = \sqrt{2}\sqrt{\cos\theta} \cdot 2\cos\theta \cdot d\theta$

$$I = 2\int_0^{\pi/2} \sqrt{2}\sqrt{\cos\theta}\; 2\cos\theta \cdot d\theta.$$

$\frac{1}{2} + 1$

$3/2$.

$$I = 4\sqrt{2}\int_0^{\pi/2} \cos^{3/2} d\theta.$$

$$\boxed{\int_0^{\pi/2} \cos^n\theta\, d\theta = \frac{\sqrt{\pi}}{2}\,\frac{\Gamma\left(\frac{n+1}{2}\right)}{\Gamma\left(\frac{n}{2}+1\right)}}$$

$$I = \cancel{4}^{\,2}\sqrt{2}\;\frac{\sqrt{\pi}}{\cancel{2}}\;\frac{\Gamma\left(\frac{\frac{3}{2}+1}{2}\right)}{\Gamma\left(\frac{\frac{3}{2}}{2}+1\right)} = 2\sqrt{2}\sqrt{\pi}\;\frac{\Gamma\left(\frac{5}{4}\right)}{\Gamma\left(\frac{7}{4}\right)}$$

$$2\sqrt{2\pi}\;\frac{\frac{1}{3}\Gamma(1/4)}{\Gamma(3/4)}$$

This fuckin Gamma Simplification

(h) $$\int_{-\pi/4}^{\pi/4} (\cos\theta + \sin\theta)^{1/3} \cdot d\theta \cdot$$

fucking shift it

$$\cos\theta + \sin\theta = \sqrt{2}\cos\left(\theta - \pi/4\right)$$

$u = \theta - \pi/4$ so $du = d\theta$.

θ	u
$-\pi/4$	$-\pi/2$
$\pi/4$	0

$$\therefore I = \int_{u=-\pi/2}^{0} (\sqrt{2}\cos u)^{1/3} du$$

$$I = (2)^{1/6} \int_{-\pi/2}^{0} (\cos u)^{1/3} du.$$

since that is an even function

$$= 2^{1/6} \int_{0}^{\pi/2} (\cos u)^{1/3} du$$

$$\int_{0}^{\pi/2} \cos^m\theta \cdot d\theta = \frac{\sqrt{\pi}}{2} \frac{\Gamma\left(\frac{m+1}{2}\right)}{\Gamma\left(\frac{m}{2}+1\right)}$$

$$(2)^{1/6} \int_{0}^{\pi/2} \cos^n\theta \cdot d\theta = \frac{\sqrt{\pi}}{2} \frac{\Gamma\left(\frac{2}{3}\right)}{\Gamma\left(\frac{7}{6}\right)} = \boxed{3 \cdot 2^{1/6} \sqrt{\pi} \frac{\Gamma(2/3)}{\Gamma(1/6)}}$$

(2) $$\int_0^1 \frac{x^2(4-x^4)}{\sqrt{1-x^2}}\,dx = 2\beta\left[\left(\frac{3}{2},\frac{1}{2}\right) - \frac{1}{2}\beta\left(\frac{7}{2},\frac{1}{2}\right)\right]$$

let $x^2 = t$.

$x = \sqrt{t}$ $\qquad dx = \frac{1}{2\sqrt{t}}\cdot dt$.

x	t
0	0
1	1

$$\therefore \frac{t(4-t^2)}{\sqrt{1-t}} \cdot \frac{dt}{2\sqrt{t}}$$

$$\frac{t^{1/2}(4-t^2)}{2\sqrt{1-t}}\cdot dt.$$

$$I = \frac{1}{2}\int_0^1 t^{1/2}(4-t^2)(1-t)^{-1/2}\cdot dt.$$

$$I = \frac{1}{2}\int_0^1 4t^{1/2} - t^{1/2+2} \quad (1-t)^{-1/2}\,dt.$$

$$I = \frac{1}{2}\int_0^1 \left(4t^{1/2} - t^{5/2}\right)(1-t)^{-1/2}\,dt.$$

$$I = \frac{1}{2}\left[\underbrace{\int_0^1 4t^{1/2}(1-t)^{-1/2}\,dt}_{M_1} - \underbrace{\int_0^1 t^{5/2}(1-t)^{-1/2}\cdot dt}_{M_2}\right]$$

$$I_1\Big|_{M_1} = 4\int_0^1 t^{3/2-1}(1-t)^{1/2-1}\cdot dt.$$

$\therefore m = 3/2$ & $n = 1/2$ $\qquad = \beta\left(\frac{3}{2},\frac{1}{2}\right)$

$$I_2|_{m_2} = \int_0^1 t^{7/2-1} (1-t)^{\frac{1}{2}-1}$$

$\therefore m = 7/2 \qquad n = 1/2$

$\therefore \beta\left(\frac{7}{2}, \frac{1}{2}\right)$

$\therefore$ upon subbing back we get $2\beta\left[\left(\frac{3}{2}, \frac{1}{2}\right) - \frac{1}{2}\beta\left(\frac{7}{2}, \frac{1}{2}\right)\right.$

3 $\displaystyle\int_0^a \frac{dx}{(a^n - x^n)^{1/n}} = \frac{\pi}{n}\ \text{cosec}\left(\frac{\pi}{n}\right)$

let $x = at$; $dx = a\,dt$.

x	t
0	0
a	1

so we need a fucking u

so: $(a^n - x^n)^{1/n} = a(1 - t^n)^{1/n}$

$$I = \int_0^1 \frac{a\,dt}{a(1-t^n)^{1/n}} = \int_0^1 \cdot(1-t^n)^{-1/n}\,dt.$$

sub 2 let $u = t^n \qquad t = u^{1/n}$

$dt = \frac{1}{n} u^{1/n - 1} \cdot du$

t	u
0	0
1	1

$$I = \int_0^1 (1-u)^{-1/n} \cdot \frac{1}{n} u^{1/n - 1} \cdot du$$

$$I = \frac{1}{n}\int_0^1 u^{1/n - 1} (1-u)^{-1/n}\,du$$

A The 2nd substitution is to get to the β fn formal which is

$$\int_0^1 x^{m-1}(1-x)^{n-1} = \beta(m, n)$$

$\therefore m = \frac{1}{n}$ & $n = 1 - \frac{1}{n}$

$\therefore I = \frac{1}{n} B\left(\frac{1}{n}, 1-\frac{1}{n}\right)$

$I = \frac{1}{n} \frac{\Gamma\left(\frac{1}{n}\right)\Gamma\left(1-\frac{1}{n}\right)}{\Gamma(1)}$

Since $\Gamma(1) = 1$ $\quad I = \frac{1}{n}\Gamma\left(\frac{1}{n}\right)\Gamma\left(1-\frac{1}{n}\right)$

now formula $\Gamma(z)\Gamma(z-1) = \frac{\pi}{\sin(\pi z)}$

$\Gamma\left(\frac{1}{n}\right)\Gamma\left(1-\frac{1}{n}\right) = \frac{\pi}{\sin\left(\frac{\pi}{n}\right)}$

$\therefore I = \frac{\pi}{n} \operatorname{Cosec}\left(\frac{\pi}{n}\right)$

4 PT $\underbrace{\int_0^{\pi/2} \sqrt{\sin\theta} \cdot d\theta}_{I_1} \cdot \underbrace{\int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin\theta}}}_{I_2} = \pi$

$\beta(p, q) = \int_0^1 t^{p-1} (1-t)^{q-1} \cdot dt.$

$\beta(p, q) = 2\int_0^{\pi/2} \sin^{2p-1}\theta \cdot \cos^{2q-1}\theta \, d\theta$

let $I_1 = \int_0^{\pi/2} \sqrt{\sin\theta} \cdot d\theta. = \int_0^{\pi/2} (\sin\theta)^{1/2} d\theta$

$2p - 1 = 1/2 \qquad 2p = 3/2 \qquad p = 3/4$ &

$2q-1=0 \quad ; \quad q = 1/2$

$$\therefore \int_0^{\pi/2} \sin^{1/2}\theta \, d\theta = \frac{1}{2}\beta(3/4, 1/2)$$

$$I_2 = \int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin\theta}} : \int_0^{\pi/2} (\sin\theta)^{-1/2} d\theta$$

$2p-1 = -\frac{1}{2} \quad ; \quad 2p = -\frac{1}{2}+1 \; ; \; 2p = \frac{-1+2}{2} ; \quad \boxed{p = \frac{1}{4}}$

$q = 1/2$

$$\therefore \int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin\theta}} \cdot d\theta = \frac{1}{2}\beta\left(\frac{1}{4}, \frac{1}{2}\right)$$

Since $I_1 \cdot I_2 = \frac{1}{2}\beta\left(\frac{3}{4}, \frac{1}{2}\right) \cdot \frac{1}{2}\beta\left(\frac{1}{4}, \frac{1}{2}\right)$

— — — — — — — — — — — — (Tedious Part)

$$\frac{1}{4}\left[\frac{\Gamma(3/4)\,\Gamma(1/2)}{\Gamma\left(\frac{3}{4}+\frac{1}{2}\right)} \cdot \frac{\Gamma(1/4)\cdot\Gamma(1/2)}{\Gamma\left(\frac{1}{4}+\frac{1}{2}\right)}\right]$$

$$\frac{1}{4}\left[\frac{\sqrt{\pi}\,\cancel{\Gamma(3/4)}}{\Gamma\frac{5}{4}} \quad \frac{\Gamma(1/4)\cdot\sqrt{\pi}}{\cancel{\Gamma(3/4)}}\right]$$

$$\cancel{\frac{1}{4}}\left[\frac{\sqrt{\pi}}{\cancel{1/4}\,\cancel{\Gamma(1/4)}} \quad \frac{\cancel{\Gamma(1/4)}\sqrt{\pi}}{}\right]$$

$\therefore \sqrt{\pi} \cdot \sqrt{\pi}$

Hence proved bitch.

$= \pi \quad = \pi$

⑤ $\underbrace{\int_0^3 \frac{x^{3/2}}{\sqrt{3-x}} \cdot dx}_{I_1} \quad \underbrace{\int_0^1 \frac{dx}{\sqrt{1-x^{1/4}}}}_{I_2} = \frac{432}{35}\pi$

Step ① never look at that collectively think of it's 2 different Problem after the 2nd Integral.

$$I_1 = \int_0^3 \frac{x^{3/2}}{\sqrt{3-x}} \cdot dx.$$

// Current goal to change $\int_0^3 \rightarrow \int_0^1$ for β fn

let $x = 3t \qquad dx = 3dt$

x	t
0	0
3	1

$$\therefore \int_0^1 \frac{(3t)^{3/2}}{\sqrt{3-3t}} \cdot 3\,dt.$$

$$= \int_0^1 \frac{3^{3/2}\, t^{3/2}}{\sqrt{3(1-t)}} 3 \cdot dt \quad ; \quad \frac{3^{3/2}}{\sqrt{3}} = 3^{3/2 - 1/2} = 3^1 = 3$$

$$I_1 = \int_0^1 \frac{3t^{3/2}}{\sqrt{1-t}} \cdot 3dt = 9\int_0^1 t^{3/2}(1-t)^{-1/2}\,dt.$$

$$\beta(p,q) = \int_0^1 t^{p-1}(1-t)^{q-1}\,dt.$$

$p - 1 = 3/2 \quad ; \quad p = \frac{3+2}{2} = 5/2$

$q - 1 = -1/2 \quad ; \quad q = \frac{-1-2}{2} = -3/2$

$$9\,\beta\left(\frac{5}{2}, \frac{-3}{2}\right) = I_1$$

$$I_2 = \int_0^1 \frac{dx}{\sqrt{1-x^{1/4}}}$$

$$= \int_0^1 (1-x^{1/4})^{-1/2}$$

Let $x^{1/4} = t$. $\quad x = t^4$

$dx = 4t^3 dt$

$$\int_0^1 4t^3 (1-t)^{-1/2} dt.$$

$p-1 = 3 \Rightarrow p = 4 \quad q-1 = -1/2 \quad q = 1/2$

$$I_2 = 4\beta(4, 1/2)$$

$$9\beta\left(\frac{5}{2} \cdot \frac{1}{2}\right) \cdot 4\beta(4, 1/2)$$

$$36\left[\frac{\Gamma(5/2)\cdot\Gamma(1/2)}{\Gamma\left(\frac{5}{2}+\frac{1}{2}\right)} \quad \frac{\Gamma(4)\,\Gamma(1/2)}{\Gamma\left(4+\frac{1}{2}\right)}\right]$$

$$36\left[\frac{\frac{3}{2}\cdot\frac{1}{2}\sqrt{1/2}\;\sqrt{\pi}}{2\times1} \quad \frac{3\cdot2\cdot1\sqrt{\pi}}{\Gamma\left(\frac{9}{2}\right)}\right]$$

$$36\left[\frac{\frac{3}{4}\pi}{2} \times \frac{6\cancel{\sqrt{\pi}}}{\frac{7}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\cancel{\sqrt{1/2}}}\right]$$

$\therefore \frac{432}{35}\pi$ Hence Proved.

⑥ $\int_0^{\pi/2} \sqrt{\tan\theta}\cdot d\theta \cdot \int_0^{\pi/2} \sqrt{\cot\theta}\cdot d\theta = \pi^2/2$

$\underbrace{\hphantom{xxxx}}_{I_2}$ $\underbrace{\hphantom{xxxx}}_{I_2}$

$$I_1 = \int_0^{\pi/2} \sqrt{\frac{\sin\theta}{\cos\theta}}\cdot d\theta = \int_0^{\pi/2} (\sin\theta)^{1/2} (\cos\theta)^{-1/2} \cdot d\theta \cdot$$

$$\int_0^{\pi/2} \sin^{2p-1}\theta \cdot \cos^{2q-1}\theta \cdot d\theta = \frac{1}{2}\beta(p, q)$$

$2p-1 = \frac{1}{2}$; $2p = \frac{3}{2}$ $p = 3/4$

$2q-1 = -1/2$; $2q = -\frac{1}{2}+1$ $2q = \frac{1}{2}$ $q = \frac{1}{4}$

$\cdot \frac{1}{2}\beta(3/4, 1/4) = I_1$

$$I_2 = \int_0^{\pi/2} \sqrt{\cot\theta}\cdot d\theta = \int_0^{\pi/2} \sqrt{\frac{\cos\theta}{\sin\theta}}\cdot d\theta$$

$$= \int_0^{\pi/2} \sin^{-1/2}\theta \ \cos^{1/2}\theta \cdot d\theta \cdot$$

$2p-1 = -1/2$; $2p = -\frac{1}{2}+1$ $p = 1/4$

$2q-1 = 1/2$; $2q = 3/2$; $q = 3/4$

$\frac{1}{2}\beta\left(\frac{1}{4}, \frac{3}{4}\right)$

$I_1 \cdot I_2 = \frac{\pi^2}{2}$

$$\frac{1}{2}\beta\left(\frac{3}{4},\frac{1}{4}\right) \cdot \frac{1}{2}\beta\left(\frac{3}{4},\frac{1}{4}\right).$$

$$\frac{1}{4}\left[\frac{\Gamma(3/4)\cdot\Gamma(1/4)}{\Gamma(1)} \cdot \frac{\Gamma(3/4)\cdot\Gamma(1/4)}{\Gamma(1)}\right]$$

$$\frac{1}{4}\left[\frac{\Gamma(3/4)\,\Gamma(1/4)}{\Gamma(1)}\right]^2$$

use the reflection formula $\Gamma(z)\cdot\Gamma(z-1) = \frac{\pi}{\sin \pi z}$

$$\boxed{\pi^2/2}$$

(7) $$\int_0^\infty \frac{e^{-x} - e^{-ax}}{x \sec x}\,dx = \frac{1}{2}\log\left(\frac{a^2+1}{2}\right).$$

$$\frac{1}{\sec x} = \cos x$$

$$I(a) = \int_0^\infty \frac{e^{-x} - e^{-ax}}{x}\cos x\,dx$$

$$\boxed{\begin{aligned} F(b) &= \int_0^\infty e^{-bx}\frac{\cos x}{x}\,dx \\ I(a) &= F(1) - F(a) \end{aligned}}$$

Frullani type integral

$$F'(b) = \frac{d}{db}\int_0^\infty e^{-bx}\frac{\cos x}{x}\,dx = -\int_0^\infty e^{-bx}\cos x\,dx$$

$$\boxed{\int_0^\infty e^{-bx}\cos x\,dx = \frac{b}{b^2+1}}$$

This is the laplace transformation of $\cos x$

$$F'(b) = -\frac{b}{b^2+1}$$

$$F'(b) = -\int \frac{b}{b^2+1}\,db = -\frac{1}{2}\ln(b^2+1)+c$$

$F(1) - F(a) \longrightarrow$ Evaluate I_s.

$$F(1) - F(a) = \left[-\frac{1}{2}\ln(1^2+1)+c\right] - \left[-\frac{1}{2}\ln(a^2+1)+c\right] = \frac{1}{2}\ln(a^2+1) - \frac{1}{2}\ln 2$$

$$\frac{1}{2}\ln\left(\frac{a^2+1}{2}\right)$$

$F(1) - F(a) = I(a)$ Proved

$$\boxed{\therefore \int_0^\infty \frac{e^{-x} - e^{-ax}}{x\sec x}\,dx = \frac{1}{2}\log\left(\frac{a^2+1}{2}\right).}$$

(8) $$\int_0^\infty e^{-\left(x^2+\frac{a^2}{x^2}\right)}\,dx = \frac{\sqrt{\pi}}{2}e^{-2a}$$

let $t = a/x$ $\quad x = a/t$ $\quad dx = -a/t^2 \cdot dt$

x	t
0	∞
∞	0

$$\therefore I = \int_\infty^0 (e)^{-(a^2/t^2+t^2)}\left(-\frac{a}{t^2}\right)dt$$

$$I = a\int_0^\infty e^{-(t^2+a^2/t^2)}\frac{dt}{t^2}$$

$$I = \int_0^\infty e^{-(x^2+a^2/x^2)}\frac{dx}{x^2} \quad \text{— (I)}$$

Then add the og integral with (I)

$$\therefore I + I = \underbrace{\int_0^\infty e^{-(x^2 + a^2/x^2)}\, dx}_{\text{The } \underline{\underline{og}}.} + \int_0^\infty e^{-(x^2+a^2/x^2)} \frac{dx}{x^2}$$

$$2I = \int_0^\infty e^{-(x^2+a^2/x^2)} \left(1 + \frac{a}{x^2}\right) dx$$

$$\frac{d}{dx}\left(x - \frac{a}{x}\right) = 1 + \frac{a}{x^2}$$

$$x^2 + \frac{a^2}{x^2} = \left(x - \frac{a}{x}\right)^2 + 2a$$

let $u = x - a/x$

$du = (1 + a/x^2) \cdot dx$

x	u
0	$-\infty$
∞	∞

$$\therefore 2I = \int_{-\infty}^{\infty} e^{-(u^2+2a)}\, du$$

$$= e^{-2a} \int_{-\infty}^{\infty} e^{-u^2} \cdot du$$

$$\int_{-\infty}^{\infty} e^{-u^2} \cdot du = \sqrt{\pi} \qquad \Rightarrow 2 \times \frac{1}{2 \cdot 2\sqrt{1}} \sqrt{\frac{I}{2}}$$

$$= \underline{\underline{\sqrt{\pi}}}.$$

$$2I = e^{-2a}\sqrt{\pi} \qquad I = \frac{\sqrt{\pi}}{2} e^{-2a}.$$

(a) $$\int_0^\infty \frac{\tan^{-1}(ax)}{x(1+x^2)}\,dx = \frac{\pi}{2}\log(1+a)$$

Double Integrals

1(a) $$\int_0^1 \int_0^y xye^{-x^2}\, dy \cdot dy$$

$$\left[\int_{y=0}^{y=1} \left[\int_{x=0}^{x=y} xye^{-x^2}\right] dx\right] \cdot dy$$

$$y \int_0^y xe^{-x^2} \cdot dx$$

let $u = x^2$

$du = 2x \cdot dx$

x	u
0	0
y	y^2

$$y \int_0^{y^2} \frac{e^{-u}}{2} du \Rightarrow \frac{y}{2} \int_0^{y^2} e^{-u} \cdot du$$

$$= y\left[-e^{-u}\right]_0^{y^2} \Rightarrow \frac{y}{2}\left(1 - e^{-y^2}\right]$$

$$\frac{y}{2}\left(1 - e^{-y^2}\right).$$

The outer Integral wrt y

$$\int_0^1 \frac{y}{2}\left(1 - e^{-y^2}\right) \cdot dy$$

$$\Rightarrow \frac{1}{2} \int_0^1 \left(y - ye^{-y^2}\right) \cdot dy$$

$$\Rightarrow \frac{1}{2}\left[\int_0^1 y\,dy - \int_0^1 y e^{-y^2}\,dy\right]$$

Let $u = y^2$

$du = 2y\,dy$

u	y
1	1
0	0

$$\Rightarrow \frac{1}{2}\left\{\left[\frac{y^2}{2}\right]_0^1 - \frac{1}{2}\left[\int_0^1 e^{-u}\,du\right]\right]$$

$$\Rightarrow \frac{1}{2}\left(\frac{1}{2} - \frac{1}{2}\left[-e^{-u}\right]_0^1\right.$$

$$\Rightarrow \frac{1}{4} - \frac{1}{4}(-e^{-1})$$

$$\Rightarrow \frac{1}{4} + \frac{1}{4}e^{-1}$$

upon combining all of those $\Rightarrow \boxed{\frac{1}{4e}.}$

(b) $$\int_0^1\int_0^1 \frac{dx\,dy}{\sqrt{1-x^2}\sqrt{1-y^2}}$$

$$\int_0^1\left[\int_{x=0}^{x=1} (1-x^2)^{1/2}(1-y^2)^{1/2}\right] dx\cdot dy$$

In this we'd be needing to seperate Integrands.

$$\frac{1}{\sqrt{1-x^2}}\cdot\frac{1}{\sqrt{1-y^2}}$$

now $\int_0^1 \frac{dx}{\sqrt{1-x^2}} \int_0^1 \frac{dy}{\sqrt{1-y^2}}$

$$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right)+C$$

$$\left[\sin^{-1}(x)\right]_0^1 \cdot \left[\sin^{-1}(y)\right]_0^1$$

$$\left\{\frac{\pi}{2}\cdot\frac{\pi}{2}\right\} = \boxed{\frac{\pi^2}{4}}$$

(c) $\int_0^{\pi/4}\int_0^{\sqrt{\cos 2\theta}} \frac{r}{(1+r^2)^2}\, dr \cdot d\theta$

$$\int_0^{\pi/4}\left[\int_{r=0}^{r=\sqrt{\cos 2\theta}} \frac{r}{(1+r^2)^2}\, dr\right] \cdot d\theta$$

let $u = 1+r^2$

$du = 2r\,dr$

r	u
0	0
$\sqrt{\cos 2\theta}$	$1+\cos 2\theta$

$1+\cos 2\theta = 2\cos^2\theta$

$$\frac{1}{2}\int_0^{2\cos^2\theta} \frac{du}{u^2} \Rightarrow \frac{1}{2} - \frac{1}{4\cos^2\theta}.$$

$$\int_{\theta=0}^{\theta=\pi/4}\left[\frac{1}{2} - \frac{1}{4\cos^2\theta}\right] \cdot d\theta.$$

$$\int_0^{\pi/2} \frac{1}{2}\theta - \frac{1}{4}\int_0^{\pi/2} \sec^2\theta \cdot d\theta$$

$$\frac{1}{2}\left[\frac{\pi}{4}\right] - \frac{1}{4}$$

So final ans :) $\dfrac{\pi - 2}{8}$.

(d) $$\int_0^{\pi/2}\int_0^{a\cos\theta} r\sqrt{a^2 - r^2}\, dr \cdot d\theta.$$

$$\int_{r=0}^{r=a\cos\theta} r\sqrt{a^2 - r^2}\cdot dr$$

$du = u = a^2 - r^2$

$\therefore du = -2r\cdot dr \Rightarrow r\,dr = -\dfrac{du}{2}$

r	u
0	a^2
$a\cos\theta$	$a^2\sin^2\theta$

$$\int_{a^2}^{a^2\sin^2\theta} u^{1/2}\left(-\frac{du}{2}\right)$$

$$-\frac{1}{2}\int_{a^2}^{a^2\sin^2\theta} u^{1/2}\,du \Rightarrow -1/2\int_{a^2}^{a^2\sin^2\theta} u^{1/2}\,du$$

$$\Rightarrow \frac{1}{2}\cdot\frac{2}{3}\left[u^{3/2}\right]_{a^2\sin^2\theta}^{a^2}$$

we changed the order of the limit so that we can remove the -ve sign.

$$\Rightarrow \frac{a^3}{3}(1 - \sin^3\theta)$$

$$\int_0^{\pi/2}\frac{a^3}{3}(1 - \sin^3\theta)\cdot d\theta$$

$$\frac{a^3}{3}\int_0^{\pi/2}(1-\sin^3\theta)\cdot d\theta$$

$$\frac{a^3}{3}\left[\int_0^{\pi/2} 1\,d\theta - \int_0^{\pi/2}\sin^3\theta\cdot d\theta\right]$$

$$\frac{a^3}{3}\left[\frac{\pi}{2} - \int_0^{\pi/2}\sin\theta\left(1-\cos^2\theta\right)d\theta\right]$$

let $t=\cos\theta$ $\quad dt = -\sin\theta\cdot d\theta$

θ	t
0	1
$\pi/2$	0

$$\int_1^0 -(1-t^2)\,dt \Rightarrow \int_0^1 (1-t^2)\cdot dt.$$

$$\Rightarrow \left[t-\frac{t^3}{3}\right]_0^1 = 1-\frac{1}{3} = \underline{\underline{\frac{2}{3}}}.$$

Plugging In.

$$I = \frac{a^3}{3}\left(\frac{\pi}{2}-\frac{2}{3}\right)$$

$$= \frac{a^3}{3}\,\frac{3\pi-4}{6}$$

$$\Rightarrow \boxed{\frac{a^3(3\pi-4)}{18}}$$

② (a) $\displaystyle\int_0^1\int_0^y xye^{-x^2}\,dx\cdot dy$

$$\int_0^1 \int_{x=0}^{x=y} x y e^{-x^2}\, dx \cdot dy$$

$$y \int_0^y x \cdot e^{-x^2} \cdot dx$$

$u = x^2 \Rightarrow du = 2x \cdot dx.$

x	u
0	0
y	y^2

$$\frac{y}{2} \int_0^{y^2} e^{-u} \cdot du$$

$$\frac{y}{2} \left[-e^{-u} \right]_0^{y^2} \Rightarrow \frac{y}{2} \left[-e^{-0} - (-e^{-y^2}) \right]$$

$$\Rightarrow \frac{y}{2} \left[1 + e^{-y^2} \right]$$

Abe Gandu tune likha ulta h!!

$$\Rightarrow \int_0^1 \left[\frac{y}{2} + \frac{y}{2} e^{-y^2} \right]$$

$$\Rightarrow \int_0^1 \frac{y}{2} dy + \int_0^1 \frac{y}{2} e^{-y^2} dy$$

$$\Rightarrow \frac{1}{2} \left[\left[\frac{y^2}{2} \right]_0^1 + \right.$$

$y^2 = u$

$du = 2y\,dy$

$$\Rightarrow \frac{1}{2}\left[\frac{1}{2} + \frac{1}{2}\int_0^1 e^{-u}\,du\right]$$

$$\Rightarrow \frac{1}{2}\left(\frac{1}{2} + \frac{1}{2}\left[-e^{-u}\right]_0^1\right]$$

$$\Rightarrow \frac{1}{4}\left[1 + \left(-e^{-1} + e^{-0}\right]\right.$$

$$\frac{1}{4}\left[2 - \frac{1}{e}\right]$$

$$\Rightarrow \left[\frac{1}{2} - \frac{1}{4e}\right]$$

(b)

$$\int_0^1 \int_0^1 \frac{dx\,dy}{\sqrt{1-x^2}\sqrt{1-y^2}}$$

$$\int_0^1 \frac{dx}{\sqrt{1-x^2}} \quad \int_0^1 \frac{dy}{\sqrt{1-y^2}}$$

$$\left[\sin^{-1}(x)\right]_0^1 \times \left[\sin^{-1}(y)\right]_0^1$$

$$\frac{\pi}{2} \times \frac{\pi}{2} \Rightarrow \boxed{\frac{\pi^2}{4}.}$$

(c)

$$\int_0^{\pi/4} \int_0^{\sqrt{\cos 2\theta}} \frac{r}{(1+r^2)^2}\, dr\, d\theta.$$

$$\int_{r=0}^{r=\sqrt{\cos 2\theta}} \frac{r}{(1+r^2)^2}\, dr$$

$u = 1+r^2$

$du = 2r\,dr$

r	u
0	1
$\sqrt{\cos 2\theta}$	$2\cos^2\theta$

$1+\cos 2\theta = 2\cos^2\theta$

$$\Rightarrow \frac{1}{2}\int_1^{2\cos^2\theta} (u)^{-2}\, du$$

$$\frac{1}{2}\int_1^{2\cos^2\theta} (u)^{-2}\, du \Rightarrow \frac{1}{2}\left[-\frac{1}{u}\right]_1^{2\cos^2\theta}$$

$$\Rightarrow \frac{1}{2}\left(-\frac{1}{2\cos^2\theta} + 1\right)$$

$$\Rightarrow \left(\frac{1}{2} - \frac{1}{4\cos^2\theta}\right)$$

$$\Rightarrow \int_0^{\pi/4} \left[\frac{1}{2} - \frac{1}{4\cos^2\theta} \right]$$

$$\Rightarrow \int_0^{\pi/4} \frac{1}{2}\, d\theta - \frac{1}{4}\int_0^{\pi/2} \sec^2\theta \cdot d\theta$$

$$\Rightarrow \frac{\pi}{8} - \frac{1}{4}\left[\tan\theta \right]_0^{\pi/4}$$

$$\Rightarrow \frac{\pi}{8} - \frac{1}{4} \quad \Rightarrow \frac{\pi - 2}{8}$$

(d) $$\int_0^{\pi/2} \int_0^{a\cos\theta} r\sqrt{a^2 - r^2}\, dr \cdot d\theta.$$

$$\int_{r=0}^{r=a\cos\theta} r\sqrt{a^2-r^2}\, dr$$

r	u
0	a^2
$a\cos\theta$	$a^2\sin^2\theta$

let $u = a^2 - r^2$

$du = -2r\,dr$

$$-\frac{1}{2}\int_{a^2}^{a^2\sin^2\theta} \sqrt{u} \cdot du \quad \Rightarrow \frac{1}{2}\int_{a^2\sin^2\theta}^{a^2} u^{1/2} \cdot du$$

$$\frac{1}{2}\left[\frac{2}{3} u^{3/2} \right]_{a^2\sin^2\theta}^{a^2}$$

$$\frac{1}{3}\left[a^3 - a^3\sin^3\theta \right]$$

$$\frac{a^3}{3}\left(1 - \sin^3\theta\right.$$

$$\int_0^{\pi/2} \frac{a^3}{3}\,(1-\sin^3\theta)\cdot d\theta \cdot$$

$$\frac{a^3}{3}\left[\int_0^{\pi/2} 1\cdot d\theta - \int_0^{\pi/2} \sin^3\theta\cdot d\theta\right]$$

$$\frac{a^3}{3}\left(\frac{\pi}{2} - \int_0^{\pi/2} \sin\theta\,(1-\cos^2\theta)\cdot d\theta\right]$$

$$= \int_0^1 (1-t^2)\cdot dt \Rightarrow \underline{\underline{2/3}}$$

$$I = \frac{a^3}{3}\left[\frac{\pi}{2} - \frac{2}{3}\right]$$

③ To change order of Integration

(a) $$\int_0^{2a}\int_{\sqrt{2ax-x^2}}^{\sqrt{2ax}} f(x,y)\,dx\cdot dy$$

$$\sqrt{2ax-x^2} \le y \le \sqrt{2ax}$$

$$y = \sqrt{2ax-x^2} \quad \& \quad y = \sqrt{2ax}$$

$$y^2 = 2ax - x^2$$

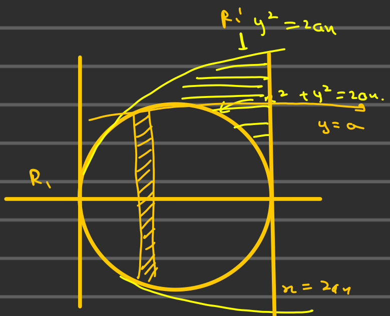
$$x^2 - 2ax - y^2$$

$$(x^2 - 2ax + a^2) - a^2 + y^2 = 0$$

$$(x-a)^2 = a^2 - y^2$$

$$x = a \pm \sqrt{a^2 - y^2}$$

x	
2a	±2a
0	0



for R_1 : $\frac{y^2}{2a} \le x \le a-\sqrt{a^2-y^2}$ $\quad 0 \le y \le a$

for R_2 $\quad \frac{y^2}{2a} \le x \le 2a$ $\quad a \le y \le 2a$

$$\int_0^a \int_{y^2/2a}^{a-\sqrt{a^2+y^2}} f(x,y)\,dx\,dy + \int_0^{2a} \int_{y^2/2a}^{4a} f(x,y)\,dx\,dy + \int_0^a \int_{a+\sqrt{a^2+y^2}}^{2a} f(x,y)\,dx\,dy$$

(b) $$\int_0^1 \int_{\sqrt{2x-x^2}}^{1+\sqrt{1-x^2}} f(x,y)\,dx\cdot dy$$

$$\int_{x=0}^{x=1} \int_{y=\sqrt{2x-x^2}}^{y=1+\sqrt{1-x^2}} f(x,y)\,dx\cdot dy$$

$x=1$ $\quad y = 1+\sqrt{1-x^2}$ $\quad (y-1)^2 = (1-x^2)$

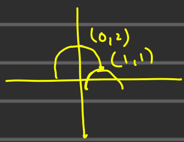
$x=0$ $\quad y = \sqrt{2x-x^2}$

if $x=0$ $\quad y = 1+1 = 2$

$x = 1$ $\quad y = 1$

$x = 2$

$y = \sqrt{2x-x^2}$



$$\sqrt{2x-x^2} \le y \le 1+\sqrt{1-x^2}$$

ⓒ $\displaystyle\int_0^{\pi}\int_0^{x} \frac{\sin y}{\sqrt{(\pi - x)(x - y}}\, dx \cdot dy$

④ change to polar coordinates

ⓐ $$\int_0^4 \int_{\sqrt{4x-x^2}}^{\sqrt{16-x^2}} \frac{dx\,dy}{\sqrt{16-x^2-y^2}}$$

$$dx\,dy = r\,dr \cdot d\theta$$

$x = 4 \quad y = \sqrt{16-x^2}$

$x = 0 \quad y = \sqrt{4x-x^2}$

Triple Integrals

① ⓐ $$\int_0^2 \int_0^x \int_0^{2x+2y} e^{x+y+z}\, dx\,dy\,dz$$

$$\int_0^2 \int_0^x \int_{z=0}^{z=2x+2y} e^x e^y e^z \, dx\,dy\,dz.$$

$$\int_0^2 \int_0^x \left[e^x \cdot e^y \int_0^{2x+2y} e^z \cdot dz \right]$$

$$\int_0^2 \int_0^x \left[e^x e^y \left[e^z \right]_0^{2x+2y} \right] dx\,dy$$

$$\int_0^2 \int_0^x \left[e^x e^y \left(e^{2x+2y} - 1 \right] dx\,dy$$

$$\int_0^2 \int_{y=0}^{y=x} \left[e^{2x} \cdot e^{2y} \cdot e^x \cdot e^y - e^x \cdot e^y \right] \cdot dx \cdot dy$$

$$\int_0^2 \int_{y=0}^{y=x} \left[e^{3x} \cdot e^{3y} - e^x e^y \right] \cdot dy \cdot dx$$

$$\int_0^2 \left[\int_0^x e^{3x} \cdot e^{3y} \cdot dy - \int_0^x e^x \cdot e^y \, dy \right] \cdot dx.$$

$$\int_0^2 \left[e^{3x} \left[3e^{3y} \right]_0^x - e^x \left(e^y \right]_0^x \right] \cdot dx$$

$$\int_0^2 \left[e^{3x}\left(3e^{3x}-3\right) - e^{x}\left(e^{x}-1\right)\right] dx$$

$$\int_0^2 \left[3e^{6x} - 3e^{3x} - e^{2x} - e^{x}\right] \cdot dx$$

$$3\int_0^2 e^{6x} \cdot dx - 3\int_0^2 e^{3x} \cdot dx - \int_0^2 e^{2x} dx - \int_0^2 e^{x} \cdot dx$$

$$18\left(e^{6x}\right]_0^2 - 9\left[e^{3x}\right]_0^2 - 2\left[e^{2x}\right]_0^2 - \left[e^{x}\right]_0^2$$

$$18\left(e^{12}-1\right) - 9\left(e^{6}-1\right) - 2\left(e^{4}-1\right) - \left(e^{2}-1\right)$$

(b) $$\int_0^1 \int_0^{1-x} \int_0^{1-x-y} \frac{1}{(1+x+y+z)^{3}} \, dx \cdot dy \, dz$$

$$\int_0^1 \int_{y=0}^{y=1-x} \int_{z=0}^{z=1-x-y} \frac{1}{(1+x+y+z)^3} \, dx \, dy \, dz.$$

$u = 1+x+y+z \qquad du = dz.$

z	u
0	1+x+y
1-x-y	2

$$\int_0^1 \int_0^{1-x} \left[\int_0^{1-x-y} u^{-3} \, dz\right] dx \, dy$$

$$\int_0^1 \int_0^{1-x} \left[-\frac{1}{2}\left(\frac{1}{u^2}\right)\right]_{1+x+y}^{2}$$

$$\int_0^1 \int_0^{1-x} -\frac{1}{2}\left[\frac{1}{4} - \frac{1}{(1+x+y)^2}\right]$$

$$\int_0^1 \int_{y=0}^{y=1-x} \frac{1}{2}\left\{\frac{1}{(1+x+y)^2} - \frac{1}{4}\right\}$$

$$\int_0^1 \left[\int_0^{1-x} \frac{1}{2}\left[\frac{1}{(1+x+y)^2} - \frac{1}{4}\right].dy\right.$$

y	u
0	1+x
1−x	2.

Let $u = 1+x+y$

$du = dy$

$$\Rightarrow \int_{1+x}^{2} u^{-2}.du \Rightarrow \left[-u^{-1}\right]_{1+x}^{2}$$

$$\Rightarrow -\frac{1}{2} + \frac{1}{1+x}$$

$$\int_{y=0}^{1-x} \frac{1}{4}\,dy = \frac{1}{4}(1-x)$$

$$\int_0^{1-x} \left(\frac{1}{(1+x+y)^2} - \frac{1}{4}\right).dy \Rightarrow \left[-\frac{1}{2} + \frac{1}{1+x}\right] - \frac{1}{4}(1-x)$$

$$\Rightarrow \frac{1}{2}\left[-\frac{3}{4} + \frac{x}{4} + \frac{1}{1+x}\right]$$

$$I = \int_0^1 \frac{1}{2}\left(-\frac{3}{4} + \frac{x}{4} + \frac{1}{1+x}\right).dx$$

$$\boxed{I = \frac{\ln 2}{2} - \frac{5}{16}}$$

ⓒ $$\int_0^{\pi/2}\int_0^{a\cos\theta}\int_0^{\sqrt{a^2-r^2}} r\cdot dz\cdot dr\cdot d\theta\cdot$$

$$\int_0^{\pi/2}\int_0^{a\cos\theta}\left[\int_{\theta=0}^{\theta=\sqrt{a^2-r^2}} r\cdot d\theta\right]\cdot dz\cdot dr$$

$$\int_0^{\pi/2}\int_{z=0}^{z=a\cos\theta} r\cdot\left[z\right]_0^{\sqrt{a^2-r^2}}\, dz\cdot dr$$

$$\int_0^{\pi/2}\left[\int_0^{a\cos\theta} r\sqrt{a^2-r^2}\cdot dz\right] dr$$

$$\int_0^{\pi/2} r\sqrt{a^2-r^2}\left(z\right]_0^{a\cos\theta} dr$$

$$a\cos\theta\int_0^{\pi/2} r\sqrt{a^2-r^2}\, dr$$

$$a^2-r^2 = u$$
$$-2r\,dr = du$$

$$\frac{a\cos\theta}{2}\int_{\pi/2}^{0}(u)^{1/2}\cdot du$$

$$\frac{a\cos\theta}{2}\int_{\pi/2}^{0}\frac{2(u)^{3/2}}{3} \Rightarrow -\frac{a\cos\theta\,(\pi/2)^{3/2}}{6}$$

(d)
$$\int_0^\infty \int_0^\infty \int_0^\infty \frac{1}{(1+x^2+y^2+z^2)} \, dx \, dy \, dz.$$

$$u = 1+x^2+y^2+z^2$$
$$du = 2x \cdot dx$$

x	u
0	0
∞	∞

$$\int_0^\infty \int_0^\infty \left[\int_0^\infty \frac{1}{u} du \right] dy \cdot dz.$$

$$\int_0^\infty \int_0^\infty \left[\log u \right]_0^\infty dy \, dz.$$

we'd be needing to use the spherical formula in here.

$$\Rightarrow \quad x = \rho \sin\theta \cdot \cos\phi$$
$$y = \rho \sin\theta \cdot \sin\phi$$
$$z = \rho \cos\theta.$$

$$1+x^2+y^2+z^2 = \rho^2 \sin^2\theta \cdot \cos^2\phi + \rho^2 \sin^2\theta \sin^2\phi + \rho^2 \cos^2\theta + 1$$
$$= \rho^2 \sin^2\theta (\cos^2\phi + \sin^2\phi) + \rho^2 \cos^2\theta + 1$$
$$= \rho^2 \sin^2\theta + \rho^2 \cos^2\theta + 1$$
$$= \rho^2 + 1$$

$$\int_0^\infty \int_0^\infty \int_0^\infty \frac{1}{(1+\rho^2)^2} dx \cdot dy \, dz.$$

$$dx \, dy \, dz = r^2 \sin\theta \, dr \, d\theta \, d\phi$$

$$\int_0^{\pi/2} \int_0^{\pi/2} \int_0^\infty \frac{\rho^2 \sin\theta}{(1+\rho^2)^2} d\rho \, d\theta \cdot d\phi$$

$$\int_0^{\pi/2} d\phi \qquad \int_0^{\pi/2} \sin\theta \cdot d\theta \qquad \int_0^{\infty} \frac{p^2}{(1+p^2)^2}\, dp$$

$$\left[\phi\right]_0^{\pi/2} \quad \left[\cos\theta\right]_{\pi/2}^{0} \to 1.$$

$$\frac{\pi}{2} \times \int_0^{\infty} \frac{p^2}{(1+p^2)^2}\, dp$$

let $p = \tan\theta \Rightarrow dp = \sec^2\theta \cdot d\theta$

p	θ
0	0
∞	$\pi/2$

$$\frac{\pi}{2} \times \int_0^{\pi/2} \frac{\tan^2\theta}{(\sec^2\theta)^2}\, dp.$$

$$\frac{\pi}{2} \times \int_0^{\pi/2} \tan^2\theta \cdot \cos^4\theta \cdot dp.$$

$$\frac{\pi}{2} \times \int_0^{\pi/2} \frac{\sin^2\theta}{\cancel{\cos^2\theta}} \times \cos^{\cancel{4}\,2}\theta \cdot dp$$

$$\frac{\pi}{2} \times \int_0^{\pi/2} \sin^2\theta \cdot \cos^2\theta \cdot dp$$

$$\frac{\pi}{2} \int_0^{\pi/2} \sin^2\theta \cdot \cos^2\theta \cdot \sec^2\theta \cdot d\theta$$

$$\frac{\pi}{2} \int_0^{\pi/2} \sin^2\theta \cdot d\theta = \pi/4. \qquad = \underline{\underline{\pi/8}}.$$

(e) $$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_{\sqrt{x^2+y^2}}^{1} \frac{1}{\sqrt{x^2+y^2+z^2}} \, dx\, dy\, dz$$

If $x^2+y^2+z^2$ always use that unit sphere formula

$x = p \sin\phi \cos\theta$
$y = p \sin\phi \sin\theta$
$z = p \cos\phi$

$$dx\, dy\, dz = p^2 \sin\phi \, dr \, d\theta \cdot d\phi$$

$$x^2+y^2+z^2 = p^2 \sin^2\phi \cos^2\theta + p^2 \sin^2\phi \sin^2\theta + p^2 \cos^2\phi$$
$$= p^2 \sin^2\phi \, (1) + p^2 \cos^2\phi$$
$$= p^2$$

$$= \frac{1}{\cancel{p}} p^{\cancel{2}} \sin\phi \, dr \, d\theta \cdot d\phi$$

$$\int_{\phi=0}^{\phi:\pi/2} \int_{\theta=0}^{\theta=\pi/2} \int_{p=0}^{p=1} p \sin\phi \, dp \, d\phi \, d\theta$$

$$\int_0^1 r\, dp = \frac{1}{2} \qquad \int_0^{\pi/2} \sin\phi \, d\phi = 1 \qquad \int_0^{\pi/2} d\theta = \pi/2$$

$$\frac{1}{2} \cdot 1 \cdot \frac{\pi}{2} = \frac{\pi}{4}$$

(f) $$\int_{x=0}^{x=1} \int_{y=0}^{y=1-x} \int_{z=0}^{z=1-x-y} x^2 y z \, dx\, dy\, dz.$$

$$\int_{z=0}^{z=1-x-y} x^2 y z \, dz$$

$$\int_0^1 \int_0^{1-x} x^2 y \left[\frac{z^2}{2}\right]_0^{1-x-y} dx\, dy$$

$$\int_0^1 \int_0^{1-x} (x^2 y - x^3 y - x^2 y^2) \cdot dy\, dx$$

$$\int_0^1 \left[\frac{x^2 y^2}{2} - \frac{x^3 y^2}{2} - \frac{x^2 y^3}{3} \right]_0^{1-x} dx$$

$$\int_0^1 \frac{x^2(1-x)}{2} - \frac{x^3 (1-x)^2}{2} - \frac{x^2(1-x)^3}{3} \cdot dx$$

$$\frac{1}{2}\int_0^1 x - x^3\, dx - \frac{1}{2}\int_0^1 x^3 (1-2x+x^2) - \int_0^1 \frac{x^2(1-3x+3x^2-x^3)\, dx}{3}$$

$$\frac{1}{2}\int_0^1 x - x^3\, dx - \frac{1}{2}\int_0^1 x^3 - 2x^4 + x^5 - \frac{1}{3}\int_0^1 x^2 - 3x^3 + 3x^4 - x^5\, dx$$

$$\frac{1}{2}\left[\frac{x^2}{2} - \frac{x^4}{4}\right]_0^1 - \frac{1}{2}\left[\frac{x^4}{4} - \frac{2x^5}{5} + \frac{x^6}{6}\right]_0^1 - \frac{1}{3}\left[\frac{x^3}{3} - \frac{3x^4}{4} + \frac{3x^5}{5} - \frac{x^6}{6}\right]_0^1$$

$$\frac{1}{2}\left(\frac{1}{2} - \frac{1}{4}\right] - \frac{1}{2}\left(\frac{1}{4} - \frac{2}{5} + \frac{1}{6}\right] - \frac{1}{3}\left[\frac{1}{3} - \frac{3}{4} + \frac{3}{5} - \frac{1}{6}\right]$$

Ans.

Q) $$\iiint \frac{1}{(1+x+y+z)^3}\, dx\, dy\, dz. \qquad \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

$x = 0$ $\qquad x/a = 1 \Rightarrow x = a$

$y = 0$ $\qquad \frac{x}{a} + \frac{y}{b} = 1 \Rightarrow y = b\left(1 - \frac{x}{a}\right)$

$z = 0$

$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \Rightarrow z = c\left(1 - \frac{x}{a} - \frac{y}{b}\right)$

$$\int_0^a \int_0^{b(1-\frac{x}{a})} \int_0^{c(1-\frac{x}{a}-\frac{y}{b})} \frac{1}{(1+x+y+z)^3} \cdot dx \cdot dy \cdot dz.$$

$$\int_0^a \int_0^{b(1-\frac{x}{a})} \int_0^{c(1-\frac{x}{a}-\frac{y}{b})} (1+x+y+z)^{-3}\, dz$$

$$\left[-\frac{1}{2(1+x+y+z)^2}\right]_0^{c(1-\frac{x}{a}-\frac{y}{b})}$$

$$\left[\frac{1}{2(1+x+y)^2} - \frac{1}{2\left(1+x+y+c\left(1-\frac{x}{a}-\frac{y}{b}\right)\right)^2}\right]$$

$$\left[\frac{1}{2(1+x+y)^2} - \frac{1}{2\left(1+x+y+c-\frac{cx}{a}-\frac{cy}{b}\right)^2}\right]$$

$$\left[\frac{1}{2(1+x+y)^2} - \frac{1}{2\left(1+c+\left(x-\frac{cx}{a}\right)+\left(y-\frac{cy}{b}\right)\right)^2}\right]$$

$$\frac{1}{2}\left[\frac{1}{(1+x+y)^2} - \frac{1}{\left(1+c+x\left(1-\frac{c}{a}\right)+y\left(1-\frac{y}{b}\right)\right)^2}\right]$$

$\alpha = 1 - \frac{a}{b}$ $\qquad \beta = 1 - \frac{y}{b}$

$$\frac{1}{2}\left[\frac{1}{(1+x+y)^2} - \frac{1}{(1+c+x\alpha+y\beta)^2}\right]$$

$$\underbrace{\int_0^{b\left(1-\frac{x}{a}\right)} \frac{1}{2}\left[\frac{1}{(1+x+y)^2}\right.}_{I_1} - \underbrace{\frac{1}{(1+c+x\alpha+y\beta)^2}}_{I_2}$$

$$\int_0^{b\left(1-\frac{x}{a}\right)} \frac{1}{(1+x+y)^2}\,dy$$

$$\left[-\frac{1}{1+x+y}\right]_0^{b\left(1-\frac{x}{a}\right)} \Rightarrow \frac{1}{1+x} - \frac{1}{1+x+b-\frac{bx}{a}}$$

$$\Rightarrow \frac{1}{1+x} - \frac{1}{1+b+x\left(1-\frac{b}{a}\right)}$$

③ $\iiint (x^2+y^2+z^2)\,dx\,dy\,dz$ $\qquad x^2+y^2+z^2=a^2$

$x = \rho \sin\phi \cos\theta$ $\qquad dx\,dy\,dz = \rho^2 \sin\theta \; d\rho\, d\theta \cdot d\phi$

$y = \rho \sin\phi \sin\theta$

$z = \rho \cos\phi$

$$(x^2+y^2+z^2) = \rho^2 \sin^2\phi \cos^2\theta + \rho^2 \sin^2\phi \sin^2\theta + \rho^2 \cos^2\phi$$
$$= \rho^2 \sin^2(\phi) + \rho^2 \cos^2\phi$$
$$= \underline{\underline{\rho^2}}.$$

$$\int_0^{\pi/2}\int_0^{\pi/2}\int_0^{a} \rho^2 \cdot \rho^2 \sin\theta \; d\phi\, d\theta\, d\rho.$$

$$\int_0^{\pi/2} d\phi \int_0^{\pi/2} \sin\theta \cdot d\theta \int_0^{a} \rho^4 d\rho.$$

$$[\phi]_0^{\pi/2} \; [-\cos\theta]_0^{\pi/2} \; \left[\frac{\rho^5}{5}\right]_0^{a}$$

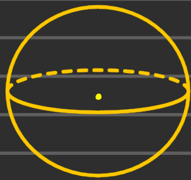
$$\frac{\pi}{2} \times 1 \times \frac{a^5}{5} \Rightarrow \underline{\underline{\frac{a^5\pi}{10}}}.$$

④ $\iiint \sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}-\frac{z^2}{c^2}}\;dx\,dy\,dz$ $\qquad \frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$

let $\frac{x}{a} = u \quad \frac{y}{b} = v \quad \frac{z}{c} = w$

$x = au \quad y = vb \quad z = wc$

$[u^2 + v^2 + w^2 = 1] \rightarrow$ This is a sphere.



$\because \{u^2+v^2+w^2 = \rho^2\}$

$\boxed{\rho = 1}$ & $dx\,dy\,dz = abc\,dw\,dv\,du$

$$I = \iiint \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2}}\,dx\,dy\,dz.$$

$$I = \iiint abc\sqrt{1-(u^2+v^2+w^2)}\,dw\,dv\,du$$

now here again we'd have to use Spherical Coordinates

$x = \rho \sin\phi \cdot \cos\theta$
$y = \rho \sin\phi \sin\theta$
$z = \rho \cos\phi$

The limits are :-

ρ :- 0 to 1
ϕ = 0 to π
θ = 0 to 2π

$dw\,dv\,du = \rho^2 \sin\phi\,d\rho\,d\phi\,d\theta$

$$\therefore I = \int_0^{2\pi}\int_0^{\pi}\int_0^{1} abc\sqrt{1-\rho^2}\,\rho^2 \sin\phi\,d\rho\,d\phi\,d\theta$$

$$I = \left(\int_0^{2\pi} d\theta\right)\left(\int_0^{\pi} \sin\phi\,d\phi\right)\left(\int_0^{1} \rho^2\sqrt{1-\rho^2}\,d\rho\right) abc$$

$$I = abc\,[\theta]_0^{2\pi}\,[-\cos\phi]_0^{\pi}\left(\int_0^{1} \frac{t(1-t)^{1/2}}{2\sqrt{t}}\,dt\right)$$

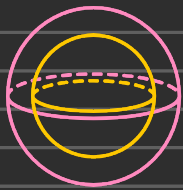
$\rho^2 = t \qquad d\rho = dt / 2\sqrt{t}.$

$$I = \frac{\pi^2 abc}{4}$$

(5) $\iiint \frac{dx\,dy\,dz}{(x^2+y^2+z^2)^{1/2}}$ $\quad \underbrace{x^2+y^2+z^2 = a^2}_{\rho = a}$ & $\underbrace{x^2+y^2+z^2 = b^2}_{\rho = b}.$

This bitch is also same as sphere i.e the limits of the Integral shou be 0 to r & 0 to π & 0 to 2π.

By spherical coordinates i.e



$x = \rho \sin\phi \sin\theta$

$y = \rho \sin\phi \cos\theta$

$z = \rho \cos\phi$

The limits are :-

$\rho : \rho : b$ to $\rho : a$

$\phi : \phi : 0$ to $\phi\ \pi$

$\theta : \theta = 0$ to $\theta = 2\pi$

& $dx\,dy\,dz = \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta$

$$x^2+y^2+z^2 = \rho^2 \sin^2\phi . \sin^2\theta + \rho^2 \sin^2\phi \cos^2\theta + \rho^2 \cos^2\phi$$
$$= \rho^2$$

$$I = \iiint \frac{dx\,dy\,dz}{(x^2+y^2+z^2)^{1/2}}$$

$$I = \int_0^{2\pi}\int_0^{\pi}\int_b^{a} \frac{1}{(\rho^2)^{1/2}} \rho^2 \sin\phi\, d\phi\, d\theta\, d\rho$$

$$I = \left(\int_0^{2\pi} d\theta\right) \times \left(\int_0^{\pi} \sin\phi \, d\phi\right) \times \left(\int_b^a \rho \, d\rho\right)$$

$$I = \left[\theta\right]_0^{2\pi} \times \left[-\cos\theta\right]_0^{\pi} \times \left[\frac{\rho^2}{2}\right]_b^a$$

$$I = (2\pi)(1+1)\left(\frac{a^2}{2} - \frac{b^2}{2}\right)$$

$$\boxed{I = 2\pi(a^2 - b^2)}$$

6 There are 2 ways to solve this problem one is formula (mentos)

$x=0, y=0, z=0$ & the plane $x+y+z = 1/6$

The intercepts are all $1/6$

The volume of tetrahedron with intercepts a, b, c is $abc/6$
here $a = b = c = 1/6$ so

$$v = \frac{1}{6} \cdot \left(\frac{1}{6}\right)^3 = \frac{1}{6} \cdot \frac{1}{216} = \frac{1}{1296}$$

The solve by triple integral method i.e

$$V = \int_{x=0}^{1/6} \int_{y=0}^{\frac{1}{6}-x} \int_{z=0}^{\frac{1}{6}-x-y} dz \, dy \, dx$$

$$= \int_0^{1/6} \int_0^{1/6 - x} \left(\frac{1}{6} - x - y\right) dy \, dx$$

After all its solving

$$v = \int_0^{1/6} \frac{1}{2}\left(\frac{1}{6}-x\right)^2 dx = \frac{1}{6} \cdot \frac{1}{216} = \boxed{\frac{1}{1296}}$$

(7) voln of tetrahedron $x=0$, $z=0$, $x=2y$ & $x+2y+z=2$

Put $x=0$ & $z=0$ in $x+2y+z=2$

$y=1$

The point is $(0,1,0)$

(2) $x=2y$ & $z=0$

$2y+2y=2$

$y=1/2$ & $x=1$

Module 6

Q1) $\int_0^1 \frac{x^2}{1+x^3}$

(i) trapezoidal rule.

$h = \frac{1-(0)}{6} = \frac{1}{6}$ → from 0 → +ve +.

x	0	1/6	2/6	3/6	4/6	5/6	1
$y = \frac{x^2}{1+x^3}$	0	0.027	0.107	0.222	0.34	0.439	0.5
	y_0	y_1	y_2	y_3	y_4	y_5	y_6

Trapezoidal rule ⇒ $I = \frac{h}{2}[\overset{(y_0+y_6)}{X} + 2\underset{\text{Rest all}}{R}]$

$I = \frac{1/6}{2}\left[(y_0+y_6) + 2(y_1+y_2+y_3+y_4+y_5)\right]$

$I \approx 0.231$

(ii) Simpson's 1/3rd rule.

$I = \frac{h}{3}[x + 2\,\text{Even} + 4\,\text{odd}]$

$I = \frac{h}{3}\left[(y_0+y_6) + 2(y_2+y_4) + 4(y_1+y_3+y_5)\right]$

$I = 0.231$

Method to find the exact value

$$\int_0^1 \frac{x^2}{1+x^3}$$

let $t = 1+x^3 \Rightarrow dt = 3x^2 dx$

$$I = \frac{1}{3}\int_1^2 \frac{dt}{t} = \frac{1}{3}\ln 2 \simeq 0.231$$

(2) $\int_0^6 \frac{dx}{1+x^2}$ using Simpson 3/8th rule; to also find error

$$h = \frac{6-0}{6} = \underline{\underline{1}}.$$

	x_0	x_1	x_2	x_3	x_4	x_5	x_6
x	0	1	2	3	4	5	6
$y = \frac{1}{1+x^2}$	1	$1/2$	$1/5$	$1/10$	$1/17$	$1/26$	$1/37$

Simpson $\frac{3}{8}$th rule $= I = \frac{3}{8}h[x + 2(\text{multiple of } 3) + 3(\text{Remaining})]$

$$I = \frac{3(1)}{8}[(y_0+y_6) + 2(x_3) + 3(x_1+x_2+x_4+x_5)]$$

$\simeq 1.35 \ldots$

$$I_{exact} = \int_0^6 \frac{dx}{1+x^2} = [\tan^{-1} x]_0^6 = \tan^{-1}(6) = 1.404$$

$$\text{error} = I_{approx} - I_{exact}.$$

$$= 1.3570 - 1.404 = -0.0485$$

(3) $\int_1^2 \frac{dx}{x}$ Simpson's 1/3rd rule taking 10 intervals

$\Rightarrow h = \frac{2-1}{10} \Rightarrow 0.1$

	x_0	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}
x	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
$y = \frac{1}{x}$	1	$1/1.1$	$1/1.2$	$1/1.3$	$1/1.4$	$1/1.5$	$1/1.6$	$1/1.7$	$\frac{1}{1.8}$	$1/1.9$	0.5

Simpson's $\frac{1}{3}$rd rule $= \frac{h}{3}[X + 2E + 4O]$

$\Rightarrow \frac{h}{3}[(y_0 + y_{10}) + 2(y_2 + y_4 + y_6 + y_8) + 4(y_1 + y_3 + y_5 + y_7 + y_9)]$

$\boxed{0.693150}$

(4) To compute $\int_{20}^{26} G(x) \cdot dx$ by i) Simpson's 1/3 rule
ii) Trapezoidal rule.

x	20	21	22	23	24	25	26
$G(x)$	95.9	96.85	97.77	96.86	99.56	100.41	101.24

Trapezoidal Rule $= \frac{h}{2}[X + 2R]$

$= \frac{1}{2}[(y_0 + y_6) + 2(y_1 + y_2 + y_3 + y_4 + y_5)]$

$= 591.81$ (Trapezoidal)

Simpson $\frac{1}{3}$rd rule $= \frac{h}{3}[X + 2E + 4O]$

$$\Rightarrow \frac{1}{3}\left[(y_0+y_6) + 2(y_2+y_4) + 4(y_1+y_3+y_5)\right]$$

$$\Rightarrow \boxed{591.8533 \text{ (Simpson } 1/3\text{rd rule)}}$$

(6) $\dfrac{dy}{dx} = \dfrac{y-x}{\sqrt{xy}}$ when $x = 1.5$ in 5 steps $\quad h = 0.1,\ y(1) = 2$

$x_0 = 1 \quad y_0 = 2$

Euler's method $y_n = y_{n-1} + hf(x_{n-1}, y_{n-1})$

$$f(x,y) = \frac{x-y}{\sqrt{xy}}$$

$x_1 = 1.1 \Rightarrow y_n = y_{n-1} + hf(x_{n-1}, y_{n-1})$

$\Rightarrow y_1 = y_0 + hf(x_0, y_0)$

$y_1 = 2 + 0.1\, f(1,2) \longrightarrow f(x,y)$ sub.

$y_1 = 2.0707$

$x_2 = 1.2 \Rightarrow y_2 = y_1 + hf(x_1, y_1)$

$y_2 = 2.0707 + 0.1\, f(1.1, 2.0707)$

$y_2 = 2.1350$

$x_3 = 1.3 \Rightarrow y_3 = y_2 + hf(x_2, y_2)$

$y_3 = 2.1350 + 0.1\, f(1.2, 2.1350)$

$y_3 = 2.1934$

$x_4 = 1.4 \Rightarrow y_4 = y_3 + hf(x_3, y_3)$

$y_4 = 2.1934 + 0.1\, f(1.3, 2.1934)$

$y_4 = 2.2463$

$x_5 = 1.5 \Rightarrow y_5 = y_4 + hf(x_4, y_4)$

$y_5 = 2.2463 + 0.1\, f(1.4, 2.2463)$

$y_5 = 2.2940$

(7) $\frac{dy}{dx} = x - y^2$ at $x = 0$ $y = 1$

$x_0 = 0, \; y_0 = 1$

$y_n = y_{n-1} + hf(x_{n-1}, y_{n-1})$ $f(x,y) = x - y^2$

$x_1 = 0.02$ $y_1 = y_0 + hf(x_0, y_0)$

$y_1 = 1 + 0.02 f(0,1)$

$y_1 = 0.98$

$x_2 = 0.04$ $y_2 = y_1 + hf(x_1, y_1)$

$y_2 = 0.98 + 0.02 f(0.02, 0.98)$

$y_2 = 0.98 + 0.02(-0.904) = 0.9611$

$x_3 = 0.06$ $y_3 = y_2 + hf(x_2, y_2)$

$y_3 = 0.9611 + 0.02 f(0.04, 0.9611)$

$y_3 = 0.9435$

$x_4 = 0.08$ $y_4 = y_3 + hf(x_3, y_3)$

$= 0.911$

(8) $\frac{dy}{dx} = xy$ $y(1) = 1$ for $x = 1.2$ $h = 0.1$

$x_0 = 1$ $y_0 = 1$

$k_1 = hf(x_0, y_0) \Rightarrow 0.1 f(1,1) = 0.1$

$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) \Rightarrow 0.1 f\left(1 + \frac{0.1}{2}, 1 + \frac{0.1}{2}\right) = 0.11025$

$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) = 0.1 f(0.1, 1.055) = 0.11708$

$k_4 = hf(x_0 + h, y_0 + k_3) = 0.1 f(1.1, 1.1107)$

$$y_1 = y_0 + \frac{1}{6}(K_1 + 2K_2 + 2K_3 + K_4)$$

y_1 is the fucking final answer.